Fundamentals of Software Systems

Intro to Data Engineering for AI Systems

Dr. Christoph Lofi

TU Delft WIS Web Information Systems


TU Delft


De Koog
Den Helder
Drachten
Sneek
Assen
A28
A32
A7
IJsselmeer
Hoogeveen
Meppel
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Alkmaar
Hoorn
A6
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Middelburg
Goes
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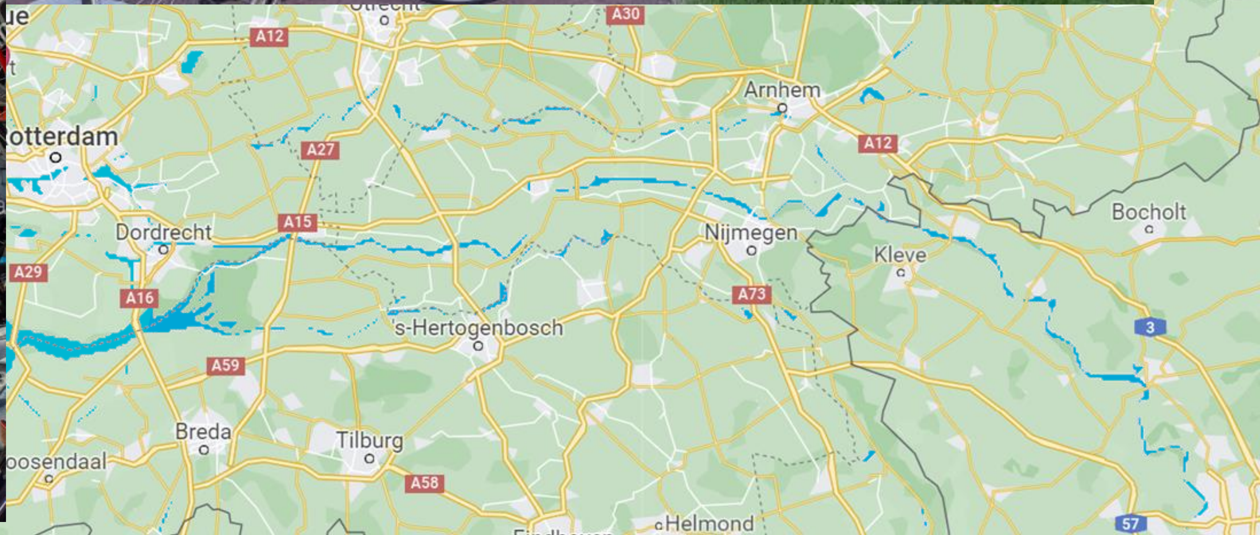
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ROYAL DELFTWARE

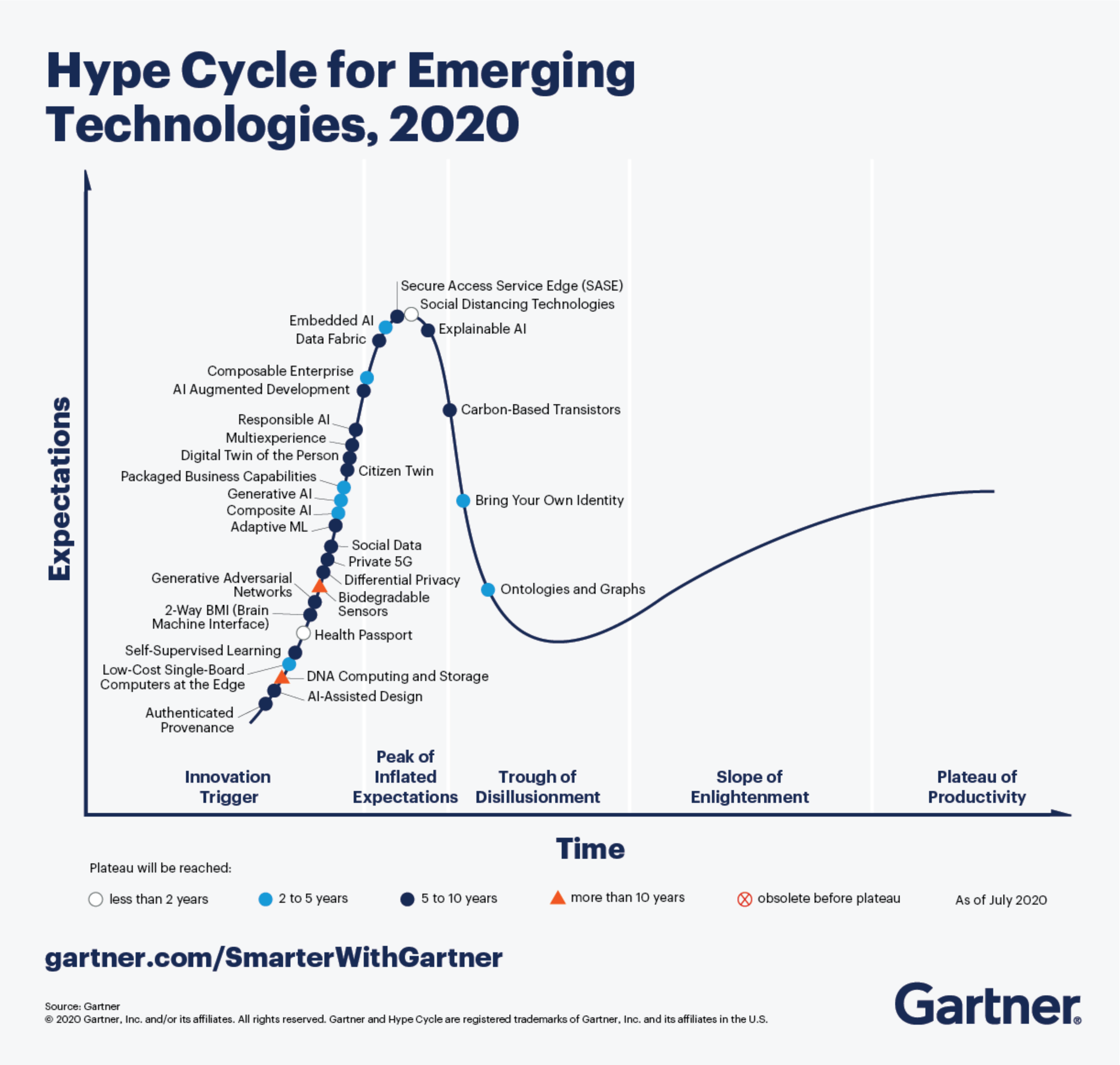
The Modern World produces a humongous amount of Data

AI is one of the Current Tech Trends

“Artificial intelligence (AI) will be the most powerful technology in generations for benefiting humanity.”

“Artificial Intelligence (AI) has the potential to solve some of **society's most important challenges**, for example the mounting pressure on **healthcare**, while at the same time enriching and improving many other aspects of our society, such as **transport**. In short, the impact of AI is enormous, and it may even become the foundation of our future.”

– from NWO LTP ROBUST


Hype Cycle for Emerging Technologies, 2020
Expectations
Secure Access Service Edge (SASE)
Social Distancing Technologies
Embedded AI
Data Fabric
Explainable AI
Composable Enterprise
AI Augmented Development
Carbon-Based Transistors
Responsible AI
Multiexperience
Digital Twin of the Person
Citizen Twin
Packaged Business Capabilities
Generative AI
Composite AI
Adaptive ML
Bring Your Own Identity
Social Data
Private 5G
Generative Adversarial Networks
Differential Privacy
Biodegradable Sensors
Ontologies and Graphs
2-Way BMI (Brain Machine Interface)
Health Passport
Self-Supervised Learning
Low-Cost Single-Board Computers at the Edge
DNA Computing and Storage
AI-Assisted Design
Authenticated Provenance
Innovation Trigger
Peak of Inflated Expectations
Trough of Disillusionment
Slope of Enlightenment
Plateau of Productivity
Time
Plateau will be reached:
less than 2 years
2 to 5 years
5 to 10 years
more than 10 years
obsolete before plateau
As of July 2020
gartner.com/SmarterWithGartner
Source: Gartner
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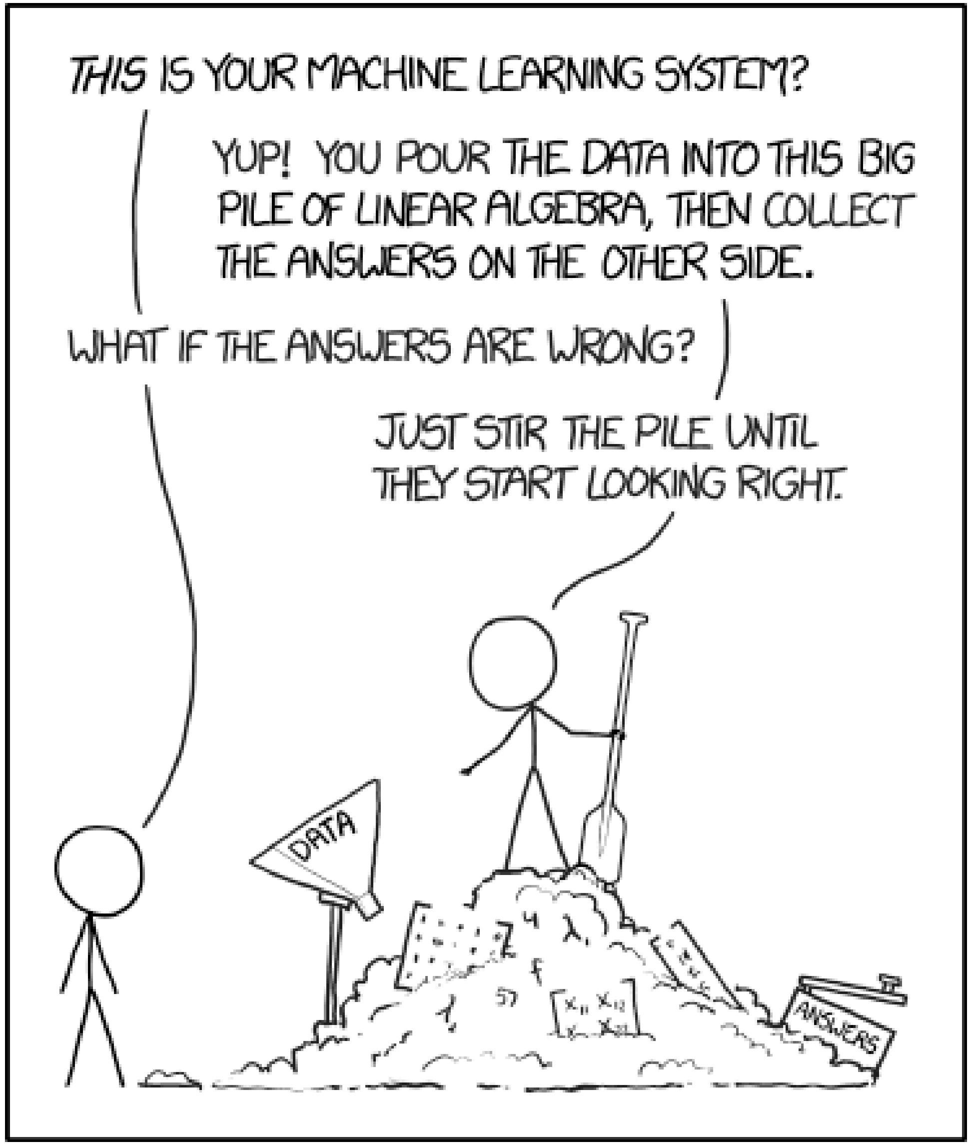
Gartner

But…

- …how successful are we really with using AI?
  - …and especially with the ever-popular Deep Learning? (DL)
- AI is not magic - it is an incredibly powerful tool for extracting regularities from data according a given objective.
  - Claim #1: **An AI system will be just as smart as the data it gets.**
  - Claim #2: **An AI system will be just as smart as the objective it optimizes.**

Challenges

- Many open challenges…
  - Where does data come from?
  - Data Quality
    - The right training data?
    - Training Data Balanced?
  - Model Evaluation
  - “Weird Features”
  - Common Sense Reasoning
  - Explanations



From xkcd.com

"To err is human' but a human error is nothing to what a computer can do if it tries."
from Agathe Christie "Hallowe'en Party", 1969

"People worry that computers will get too smart and take over the world, but the real problem is that they're too stupid and they've already taken over the world."
from Pedro Domingos "The Master Algorithm", 2015

The "hidden depth" of ML

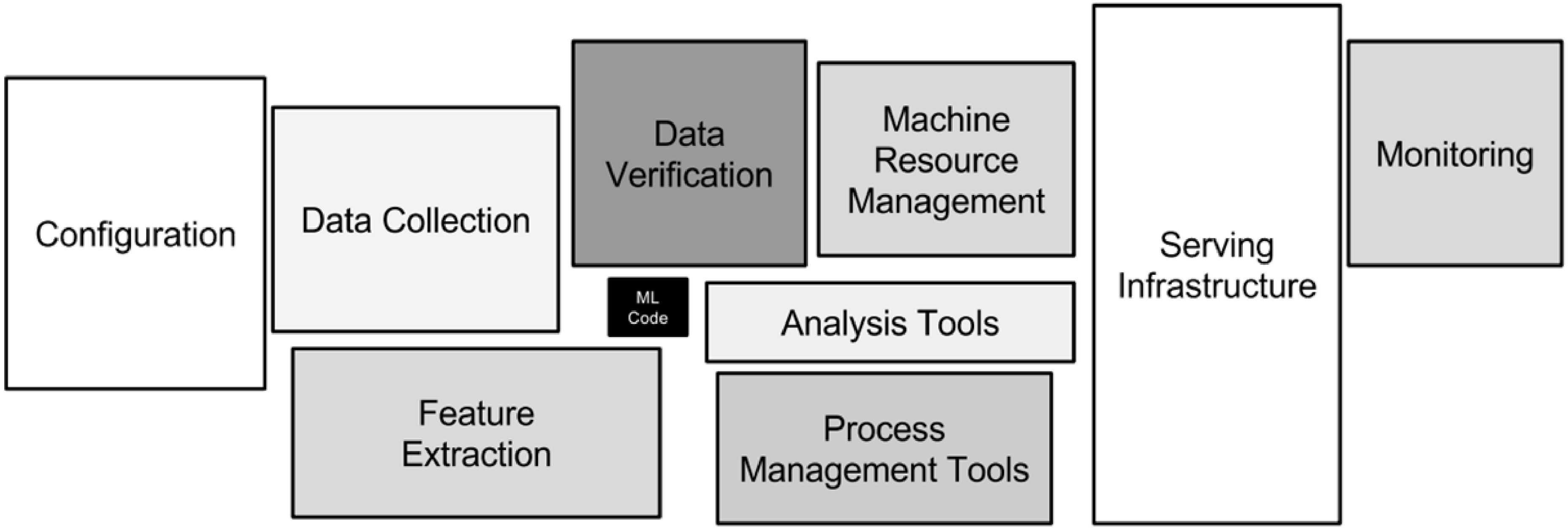
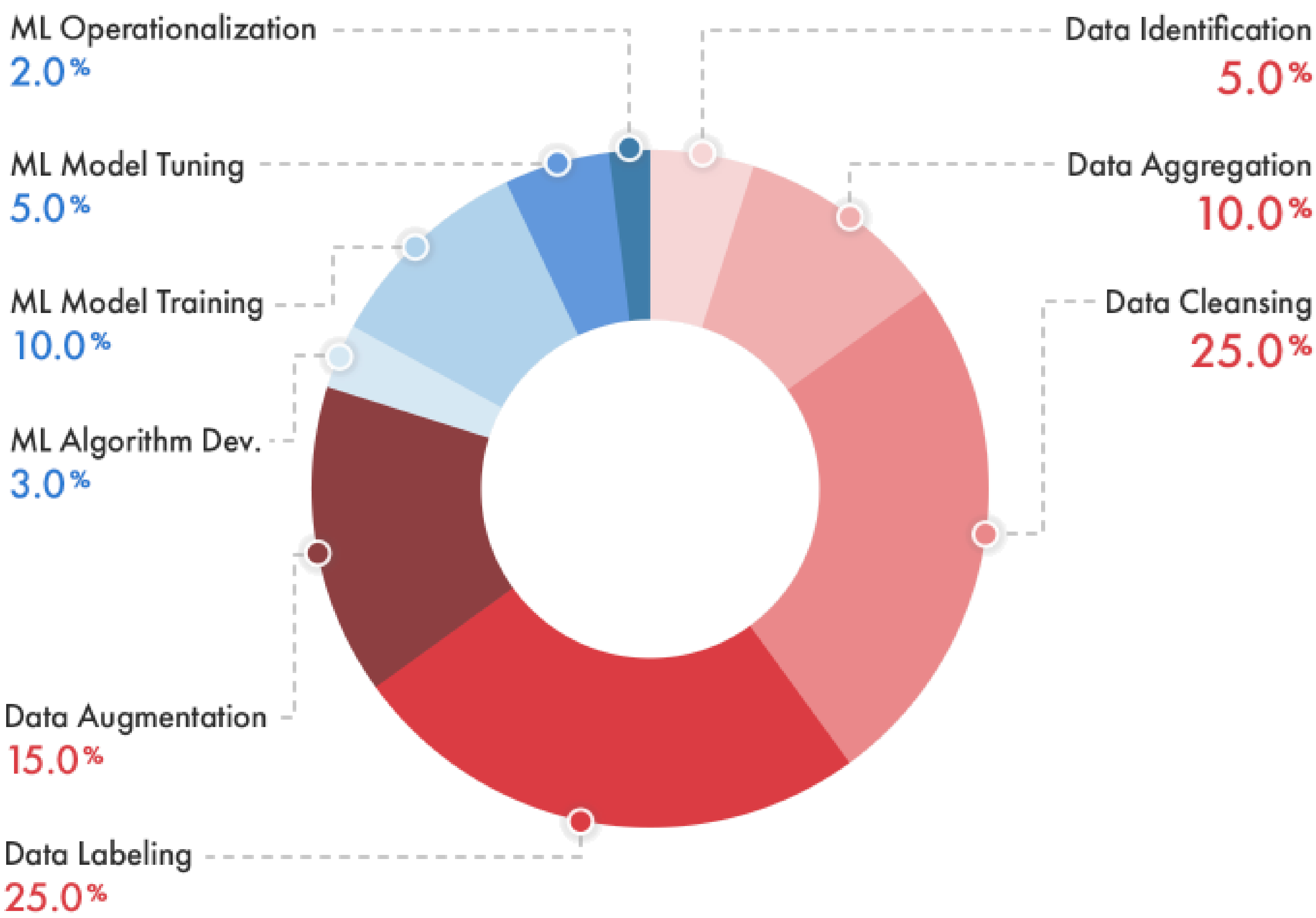


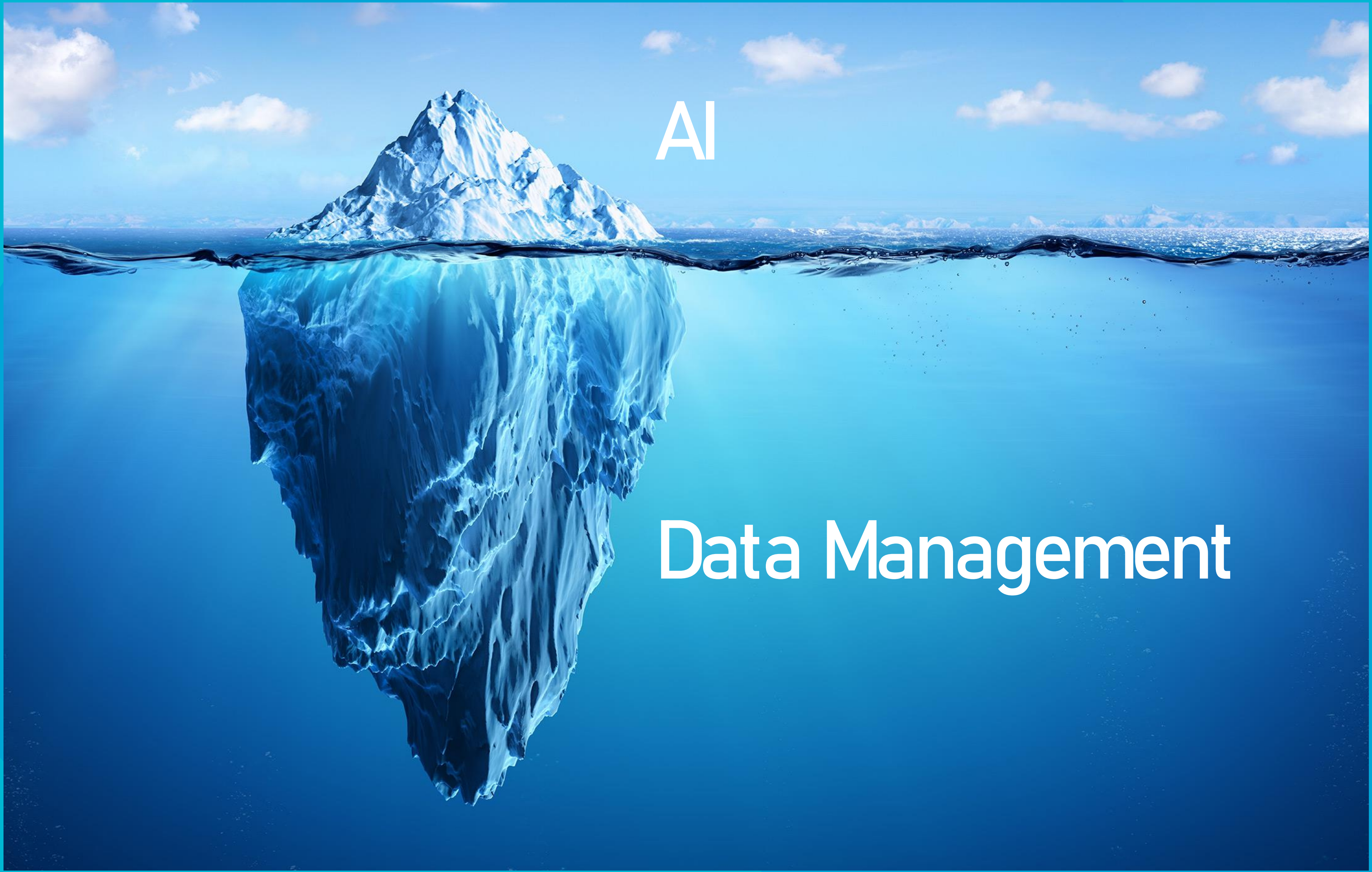
Figure 1: Only a small fraction of real-world ML systems is composed of the ML code, as shown by the small black box in the middle. The required surrounding infrastructure is vast and complex.

Hidden Technical Depth of Machine Learning, D. . Sculley, NIPS 2015

Percentage of Time Allocated to Machine Learning Project Tasks



https://towardsdatascience.com/data-labeling-is-chinas-secret-weapon-in-the-connected-car-battle-e8e395965380


AI
Data Management

AI and Big Data

Putting Big Data to use?

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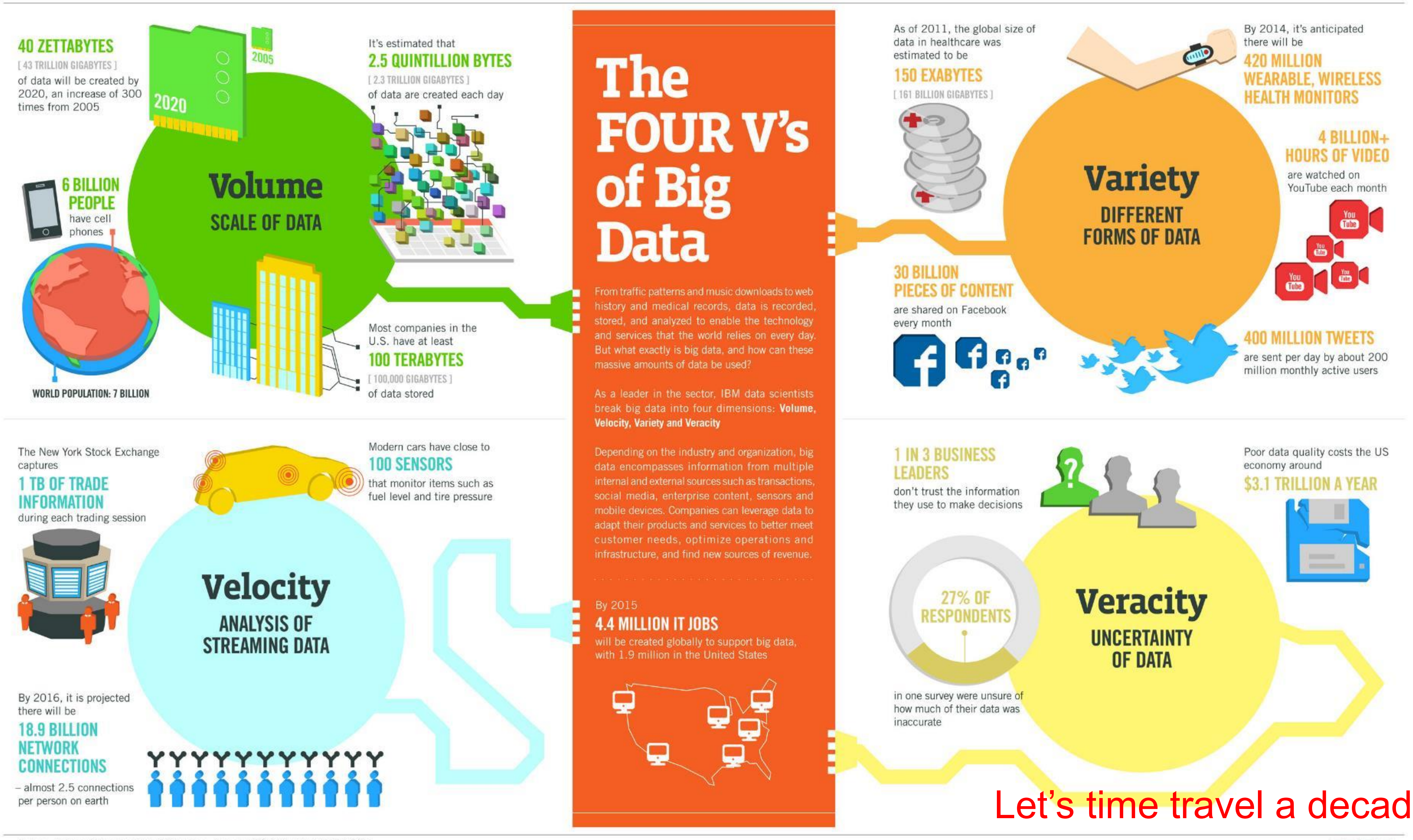
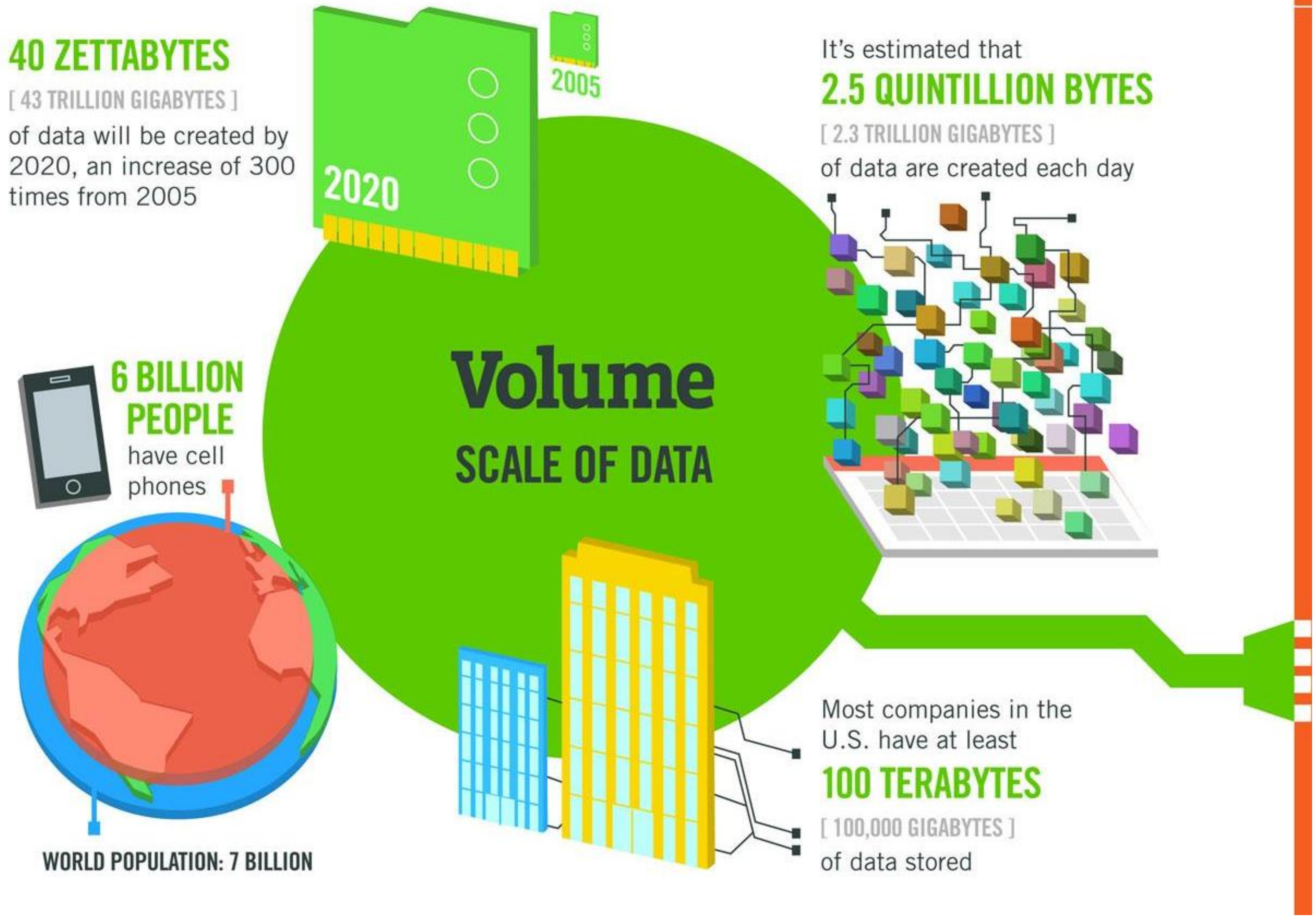
“All you’re seeing now — all these feats of AI like self-driving cars, interpreting medical images, beating the world champion at Go and so on — these are very narrow intelligences, and they’re really trained for a particular purpose. They’re situations where we can collect a lot of data.”

Yann LeCun, Facebook Head of AI, 2017

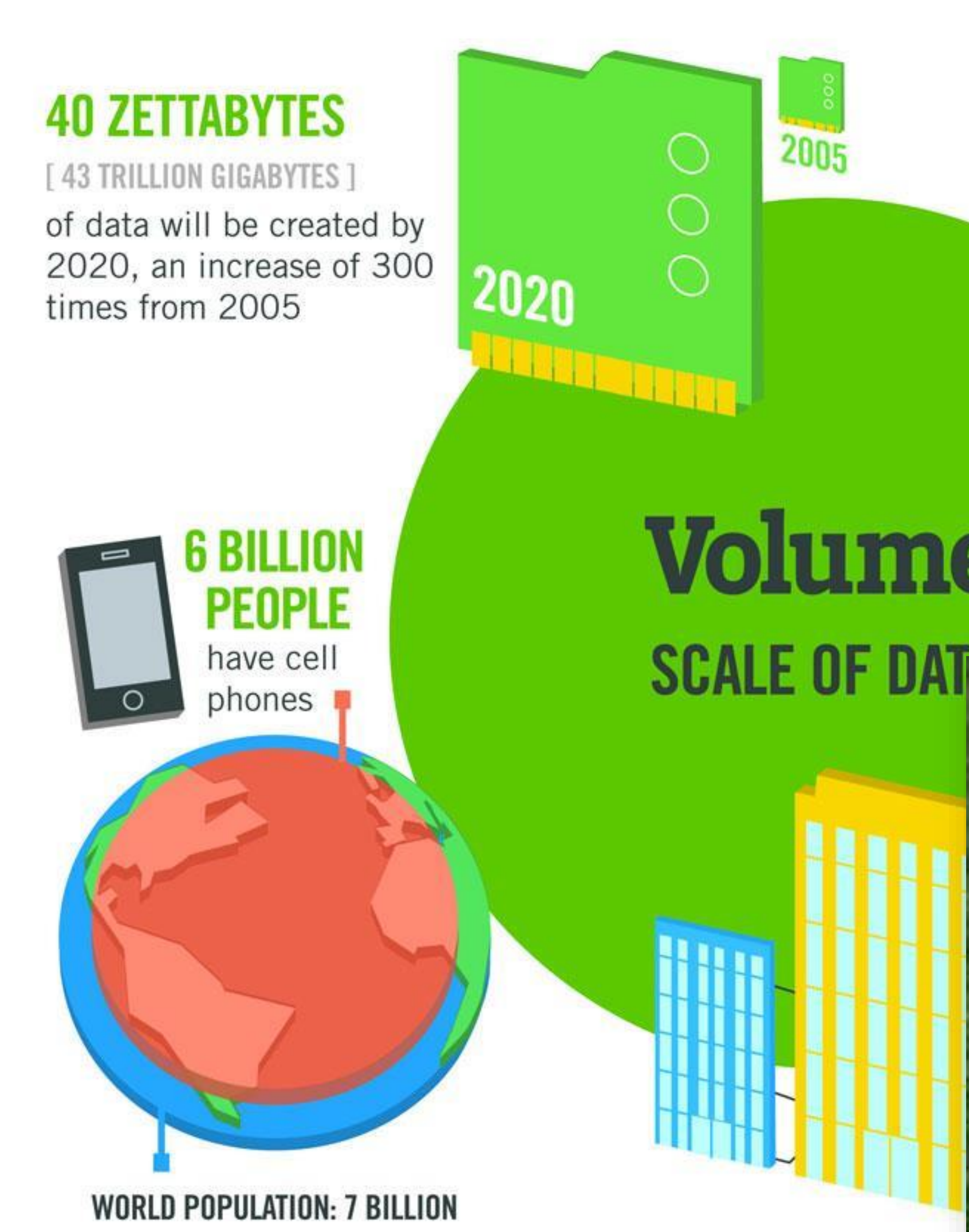


Let's time travel a decade..

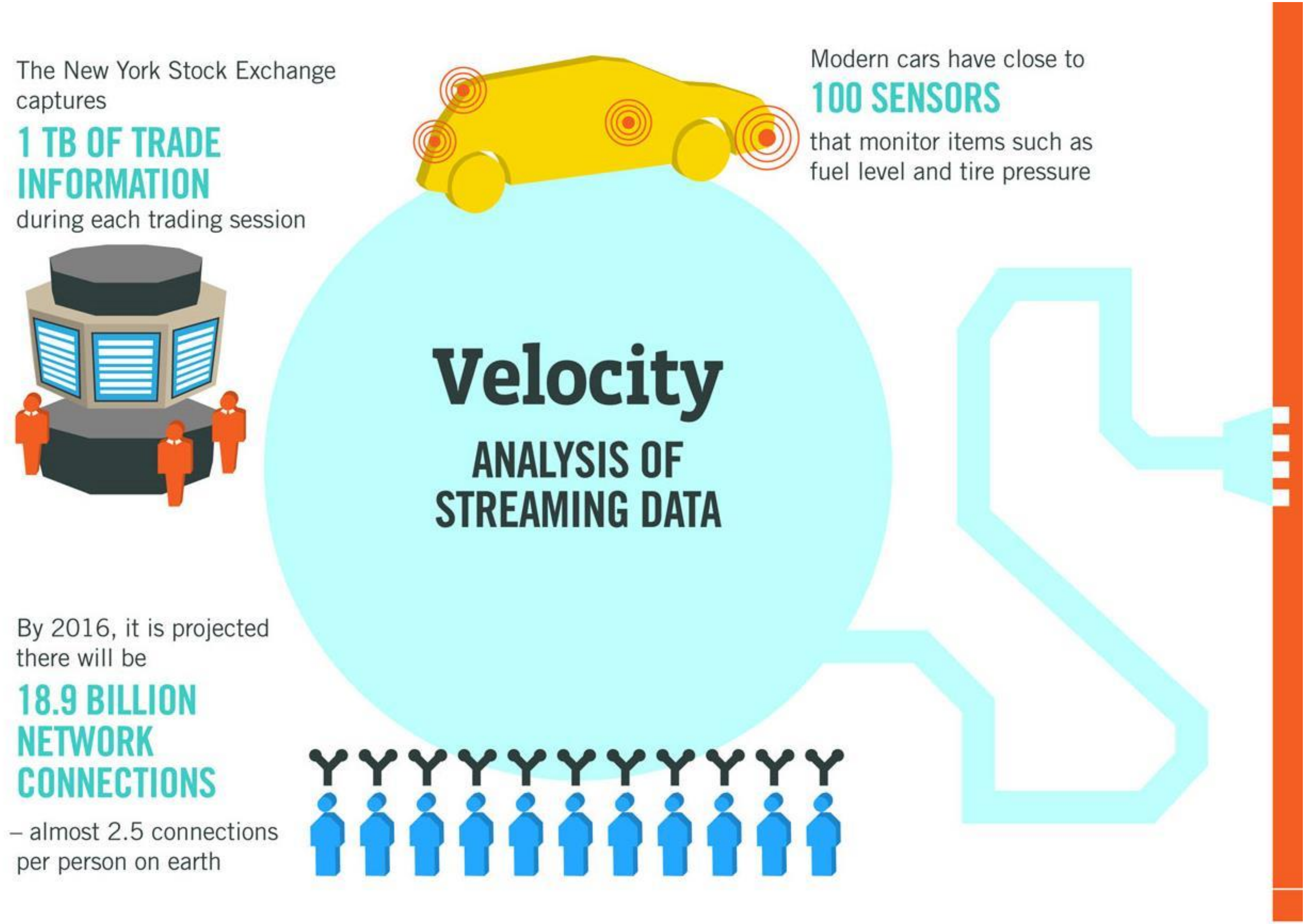
From IBMs Big Data Hub: http://www.ibmbigdatahub.com/sites/default/files/infographic_file/4-Vs-of-big-data.jpg

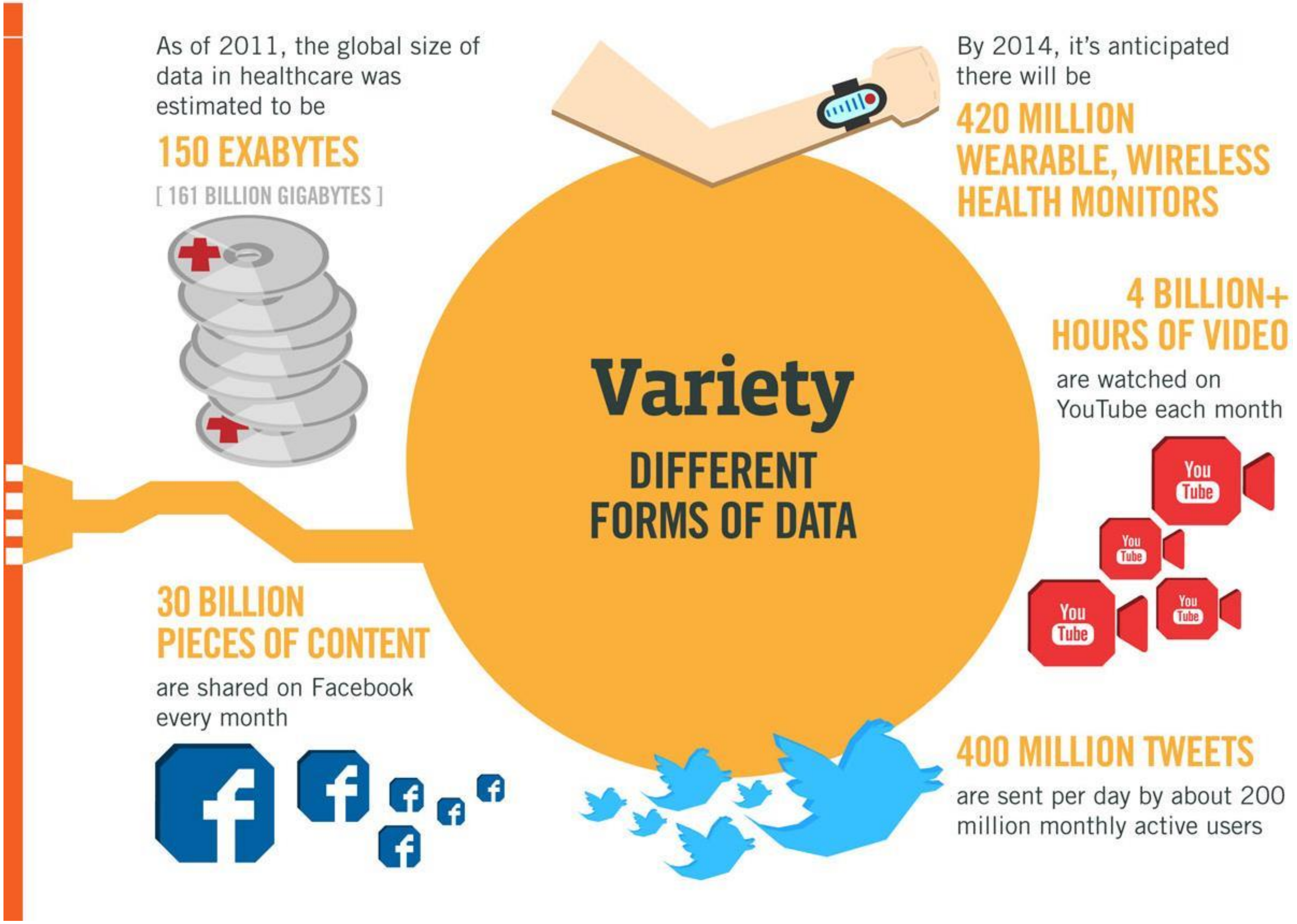


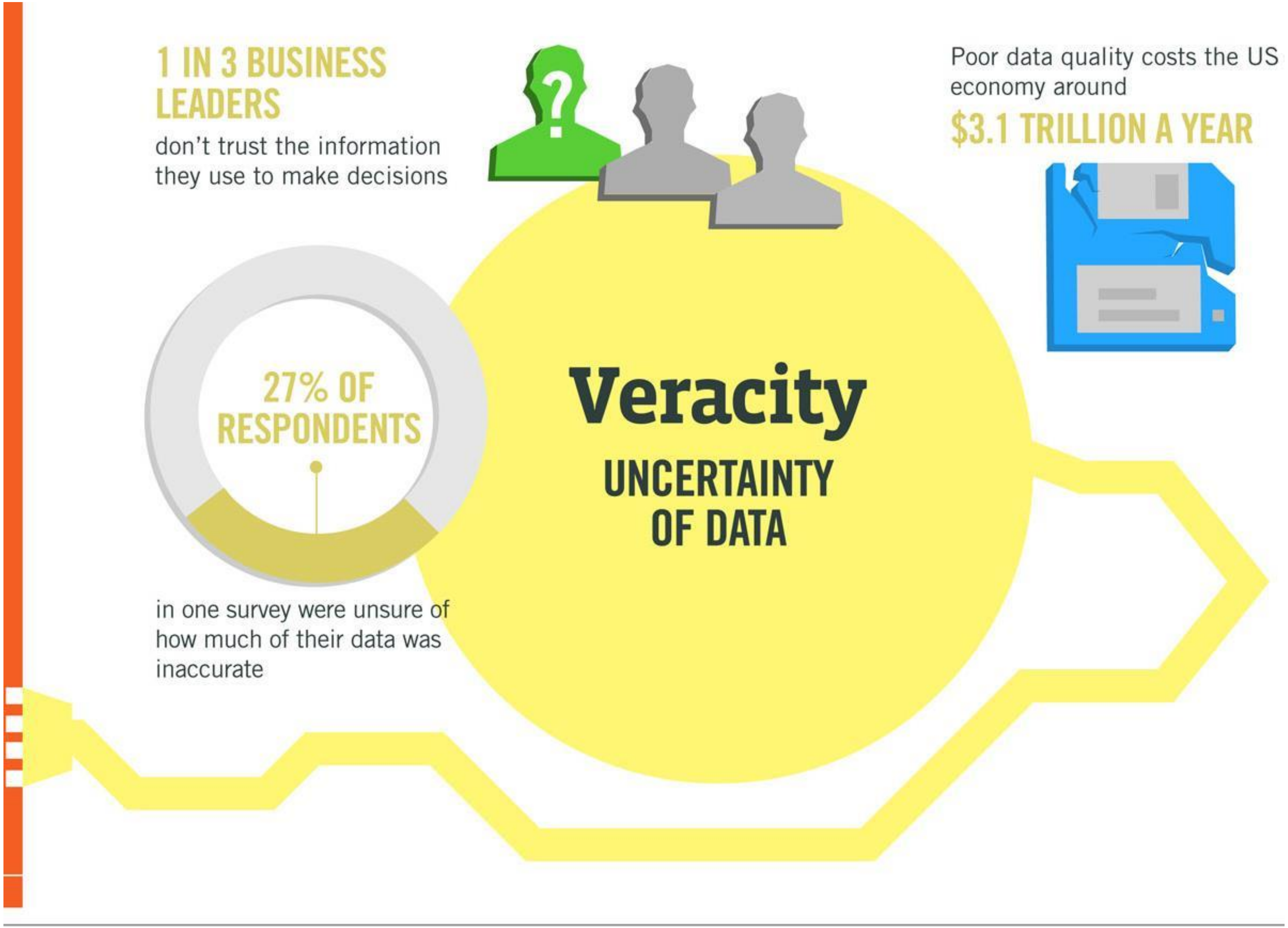
TU Delft Web Information Systems WIS


40 ZETTABYTES
[43 TRILLION GIGABYTES]
of data will be created by 2020, an increase of 300 times from 2005
2020
2005
It's estimated that
2.5 QUINTILLION BYTES
[2.3 TRILLION GIGABYTES]
of data are created each day
6 BILLION PEOPLE
have cell phones
Volume
SCALE OF DAT
WORLD POPULATION: 7 BILLION


WHARRGARBL


The New York Stock Exchange captures
1 TB OF TRADE INFORMATION
during each trading session
Modern cars have close to
100 SENSORS
that monitor items such as fuel level and tire pressure
Velocity
ANALYSIS OF STREAMING DATA
By 2016, it is projected there will be
18.9 BILLION NETWORK CONNECTIONS
– almost 2.5 connections per person on earth

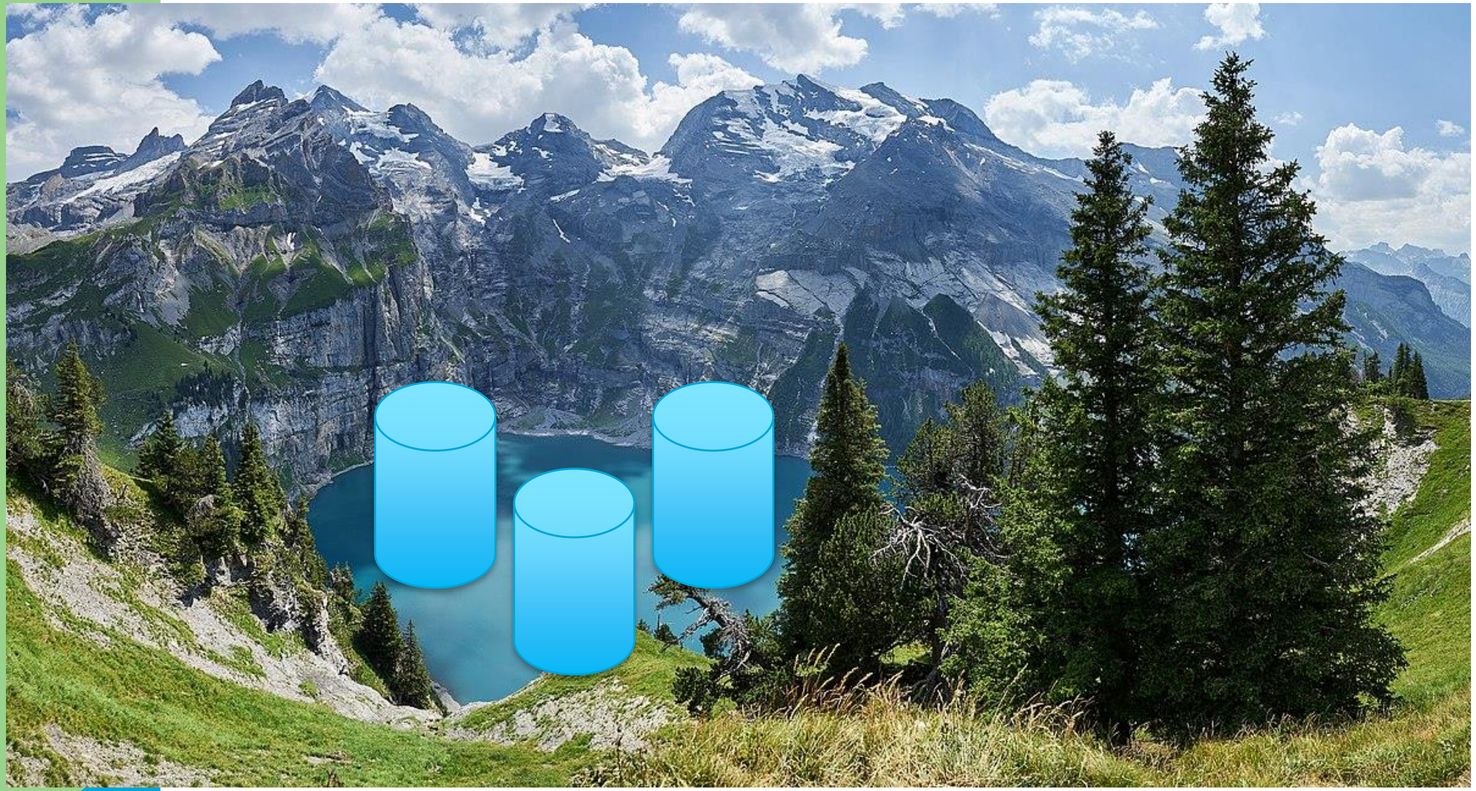

As of 2011, the global size of data in healthcare was estimated to be
150 EXABYTES
[161 BILLION GIGABYTES]
By 2014, it's anticipated there will be
420 MILLION WEARABLE, WIRELESS HEALTH MONITORS
Variety
DIFFERENT FORMS OF DATA
4 BILLION+ HOURS OF VIDEO
are watched on YouTube each month
30 BILLION PIECES OF CONTENT
are shared on Facebook every month
400 MILLION TWEETS
are sent per day by about 200 million monthly active users


1 IN 3 BUSINESS LEADERS
don't trust the information they use to make decisions
Poor data quality costs the US economy around
$3.1 TRILLION A YEAR
27% OF RESPONDENTS
in one survey were unsure of how much of their data was inaccurate
Veracity
UNCERTAINTY OF DATA

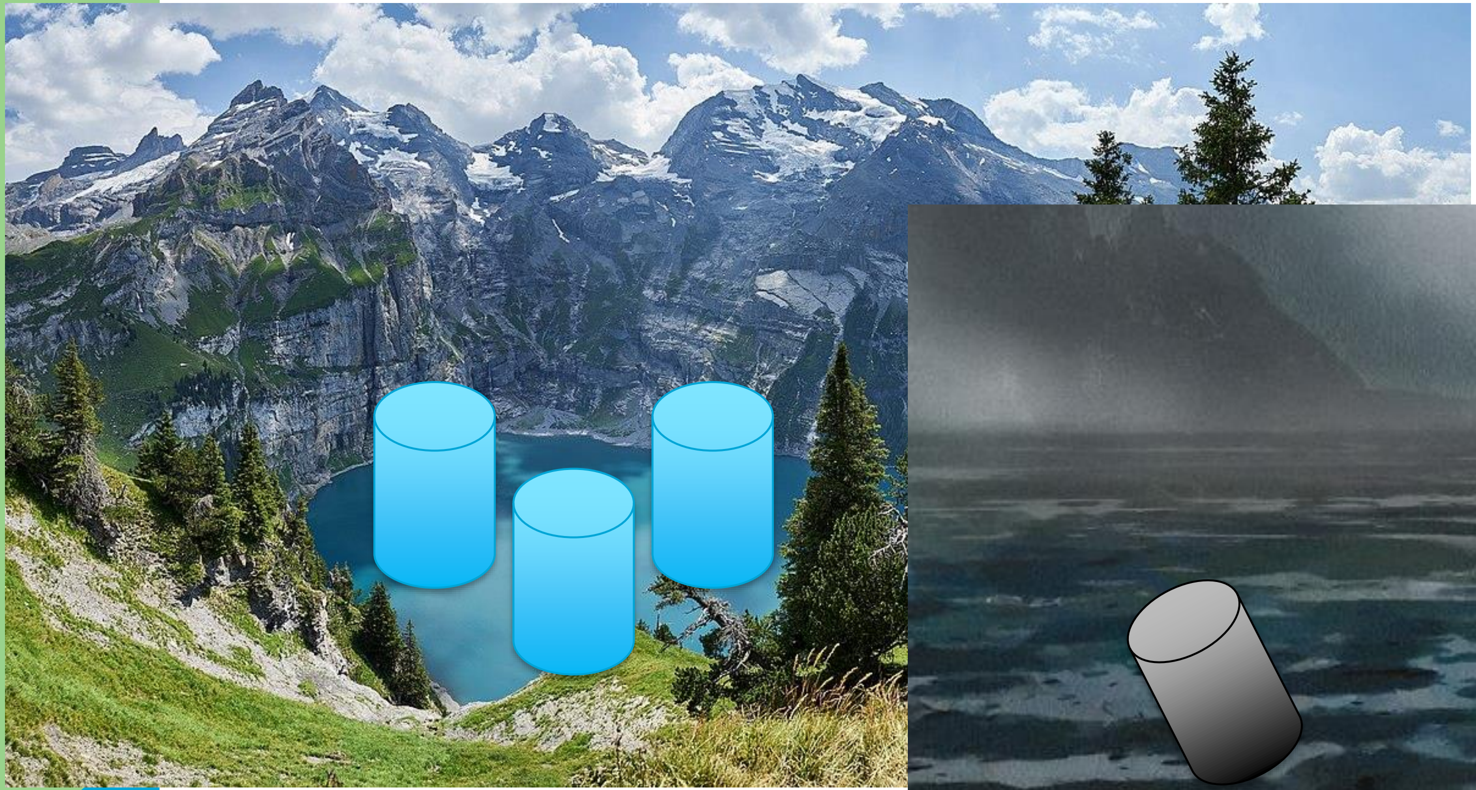
Semantics-Based Data Engineering

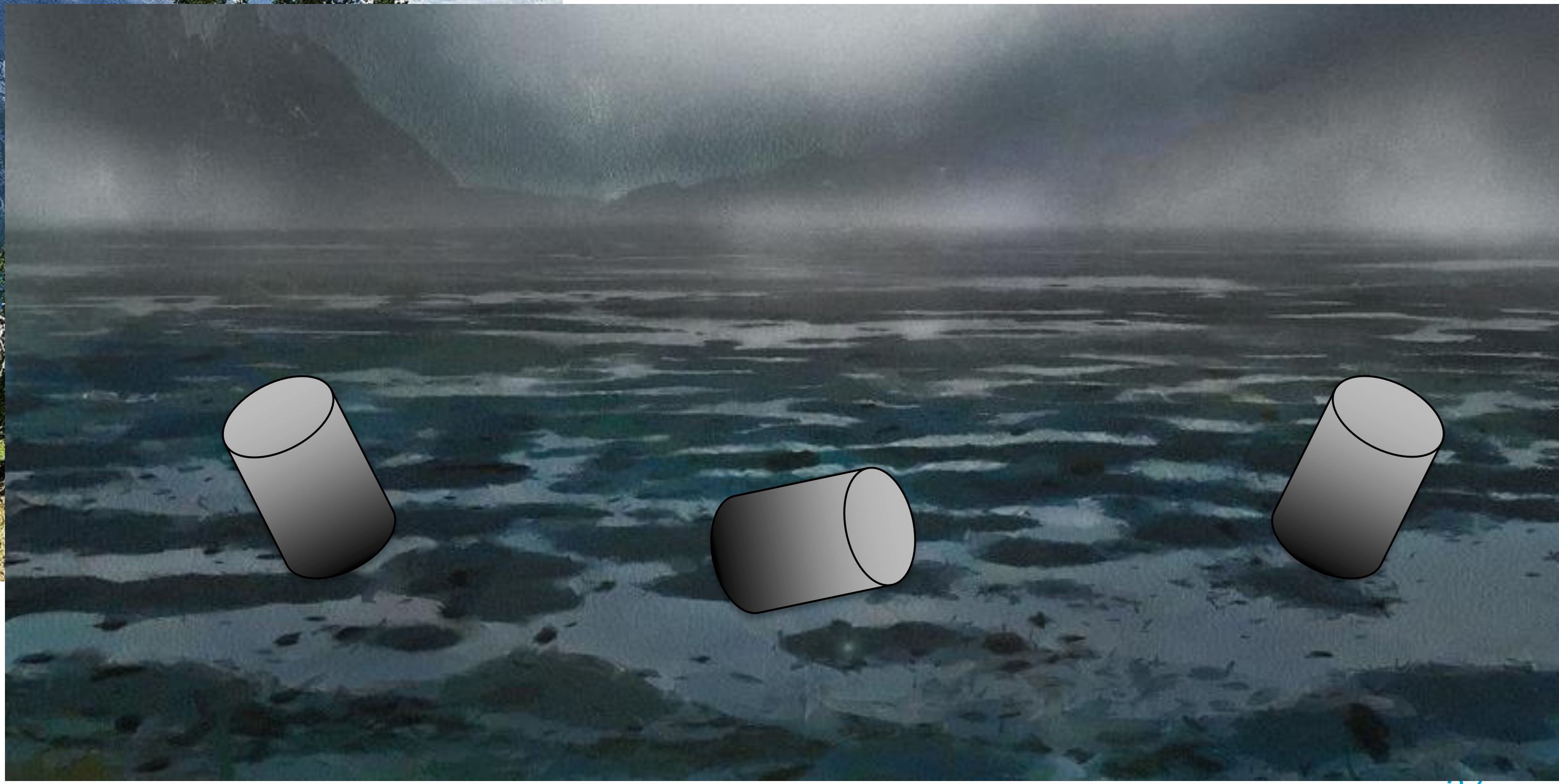
- Big Variety
  - Source Data: Unstructured implicit semantics
  - Target Data: Structured explicit semantics
- Big Veracity
  - Can we trust our input data?
  - Is it useful for the task?

Data Lakes?



Data Lakes? Or Data Swamp?





Big Variety – Data Integration

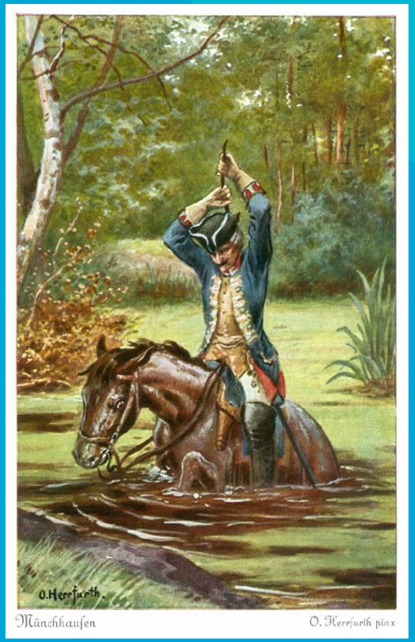
- The core challenge for Big Data Systems:
  - Data INTEGRATION
    - "Big Variety is the 800-pound gorilla in the corner."
      - Michael Stonebreaker, during Touring Awards 2015
  - Data CLEANING
    - "The 80/20 data science dilemma"

Big Veracity

- Core challenge here:
  - Is my data correct?
  - Is my data good?
  - Is my data useful for the purpose?
  - Can I actually answer any of the above questions?

Where does all my data come from?

Make it, buy it, steal it, fabricate it…



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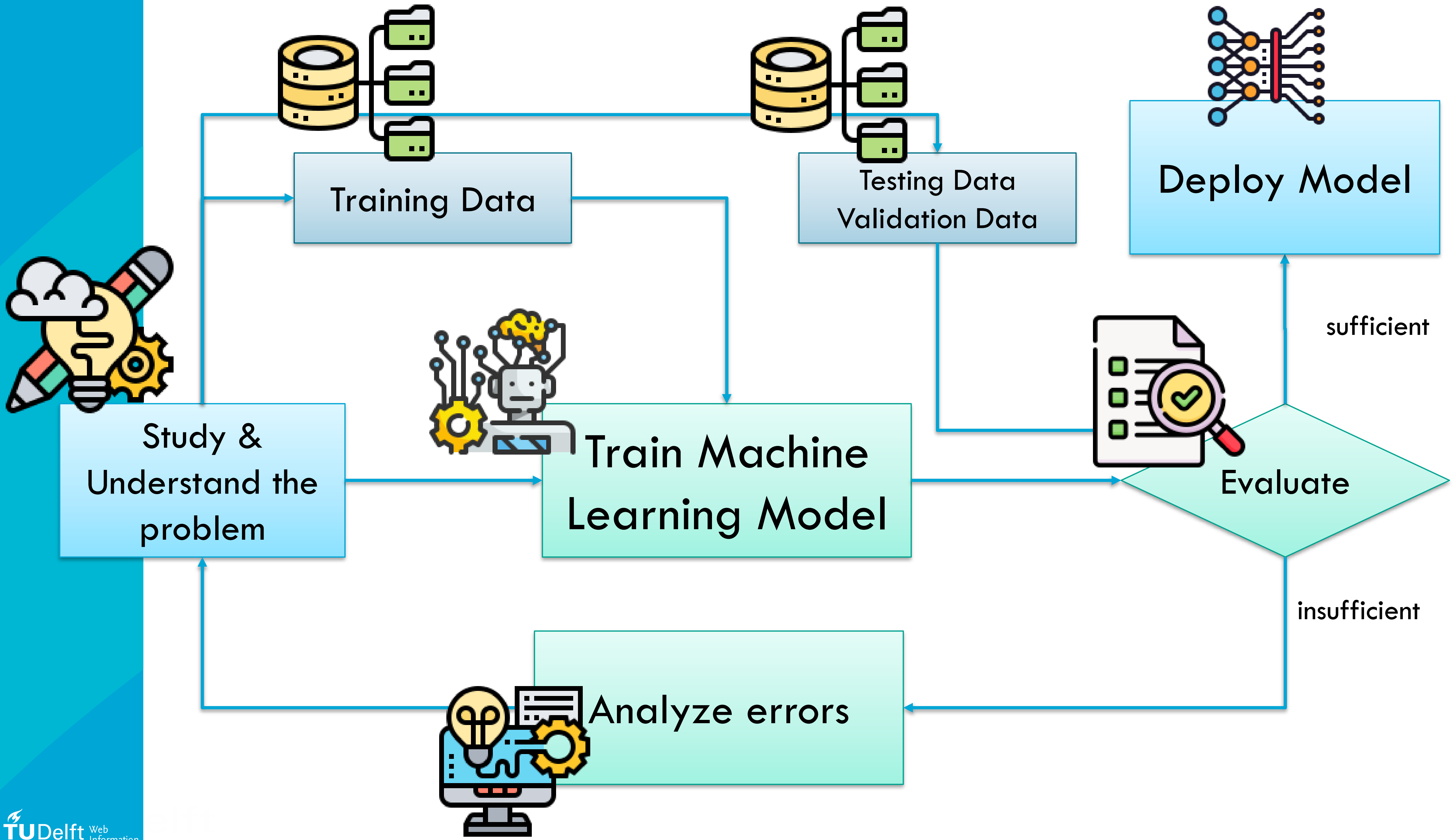
Training Data
Testing Data
Validation Data
Deploy Model
sufficient
Study &
Understand the
problem
Train Machine
Learning Model
Evaluate
insufficient
Analyze errors

Where does ML data come from?

1. Reuse Open ML Training Data
   – Platforms like Kaggle, or your university course providers
2. Data Repositories and Data Marts
   – “Buy” or “Get” data
3. Your own Enterprise Database Management System
   – “Make your own data”
4. Data Farms / Crowd Sourcing
   – Use (and pay?) a crowd of humans to make the data for you
5. Artificial/Synthetic Data Training Data
   – “Use AI-generated data to train another AI”
6. “Steal It”
   – Crawl data from everywhere without any consideration about rights and wrong whatsoever....

1 Reuse Open ML Training Data

- Platforms like Kaggle, or your university course providers
- These datasets are typically tailor-made for a specific machine learning task!
  - Are they relevant to YOUR problem?

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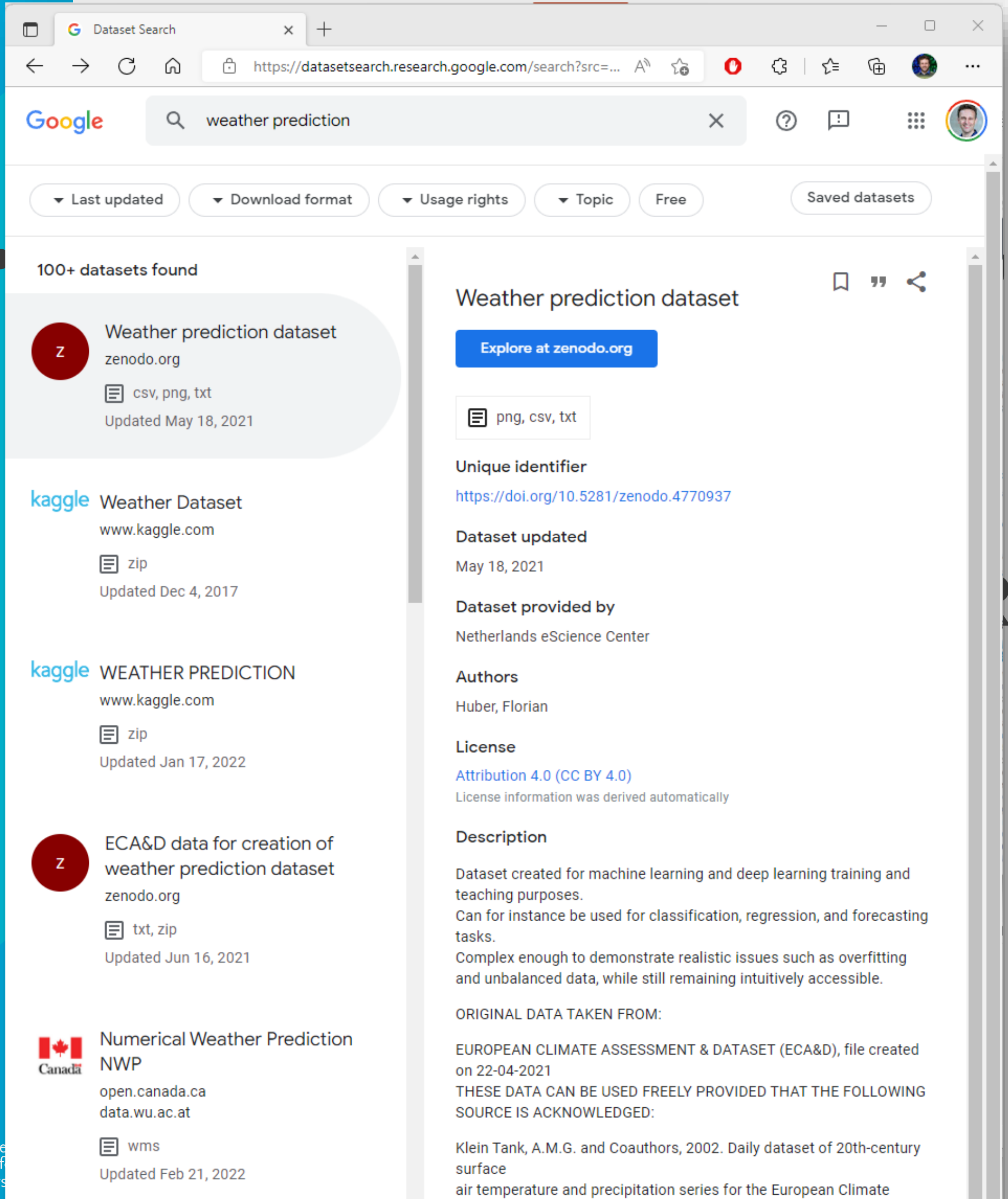
//www.kaggle.com/datasets
h: https://datasetsearch.research.google.com/
gistry: https://registry.opendata.aws/

1 Reuse Open ML Traini



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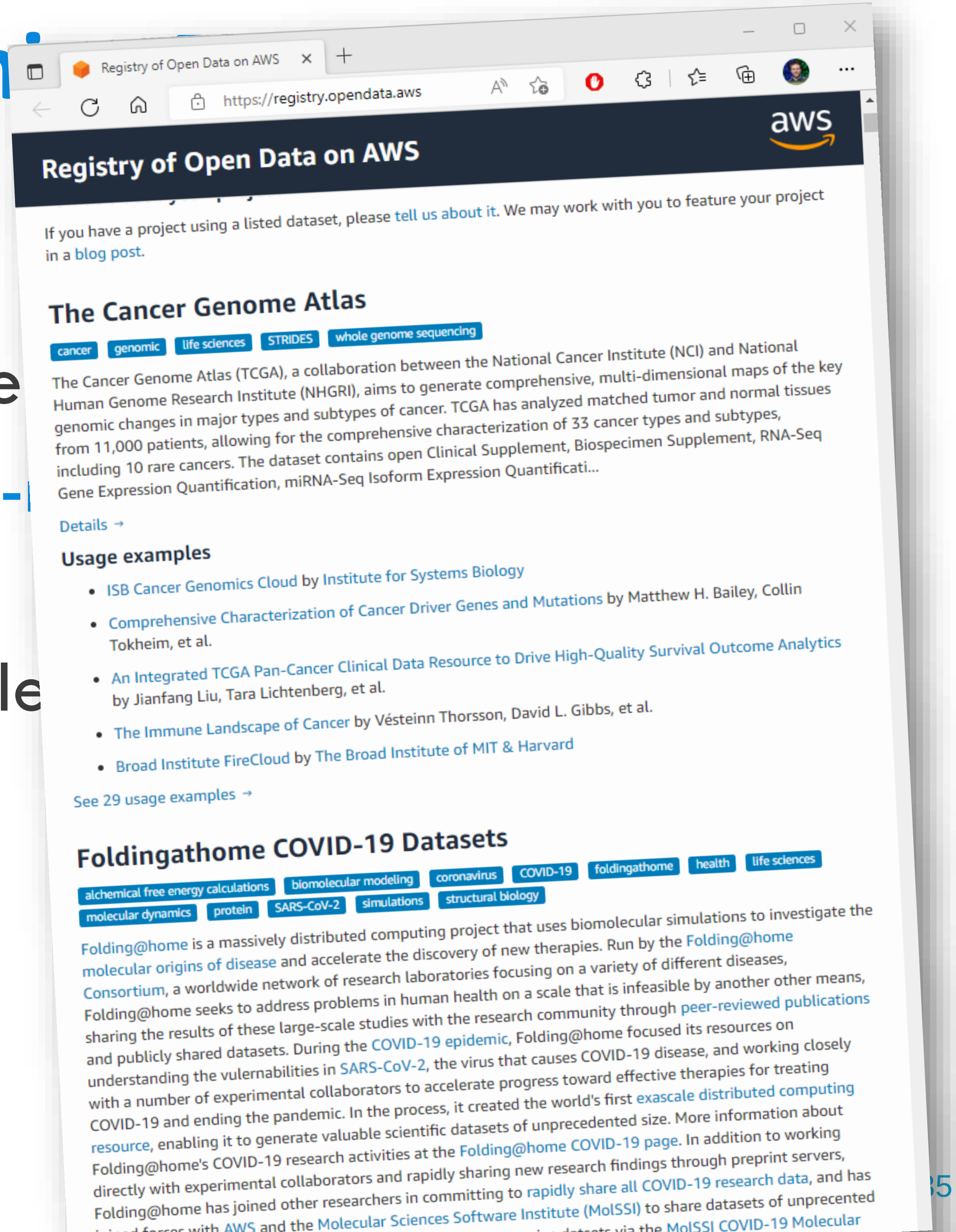

2. Data Repositories and Data Marts

- “Buy” or “Get” data
- Many companies and institutions collect large datasets and provide them for a fee (or sometimes for free)
- Often, these datasets are not made specifically for ML training, and to augmented before use (like adding labels)

https://opendata.cbs.nl/#/CBS/nl/

eid Data: https://data.overheid.nl/en/datasets

https://opendata.cbs.nl/#/CBS/nl/
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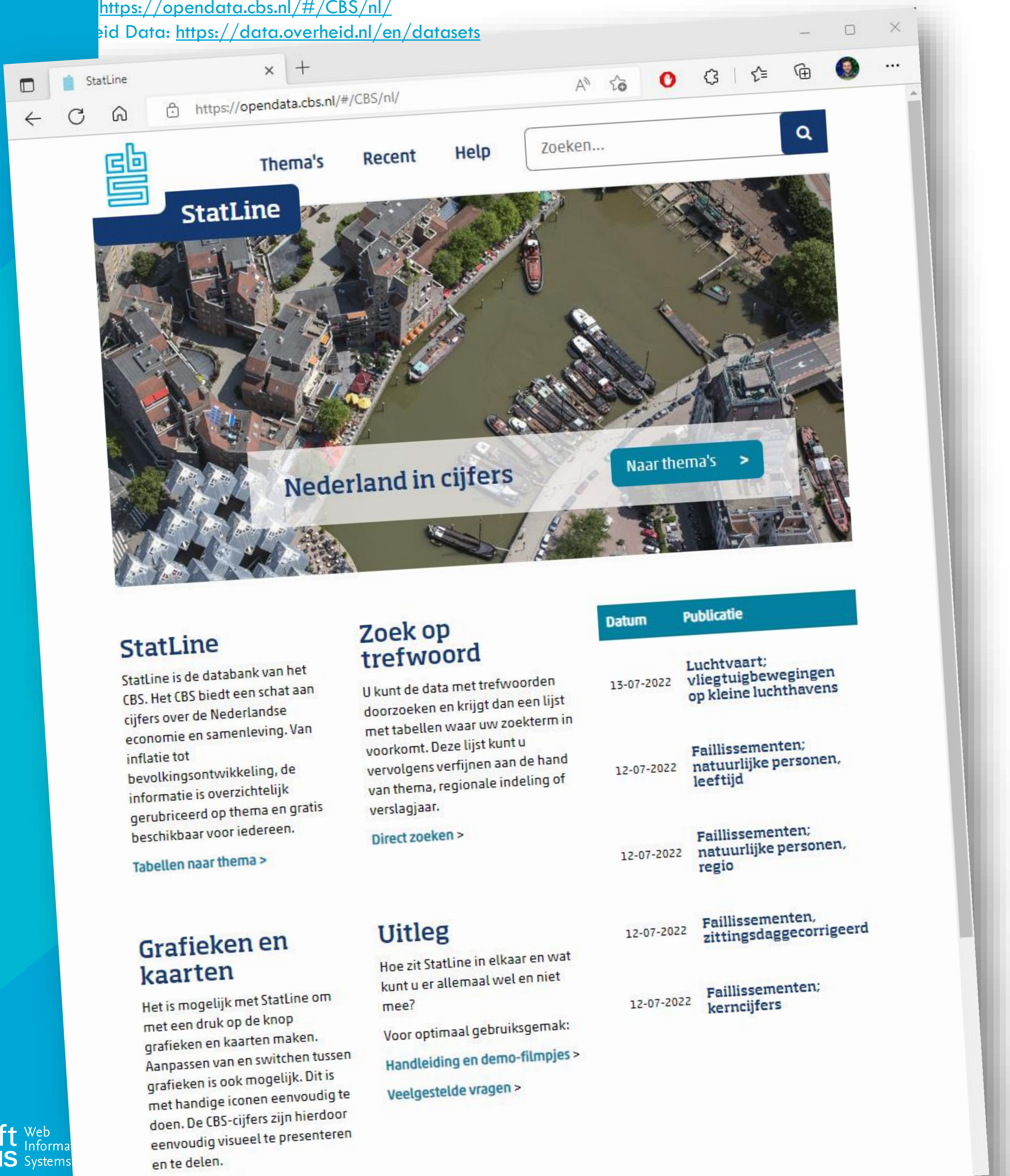
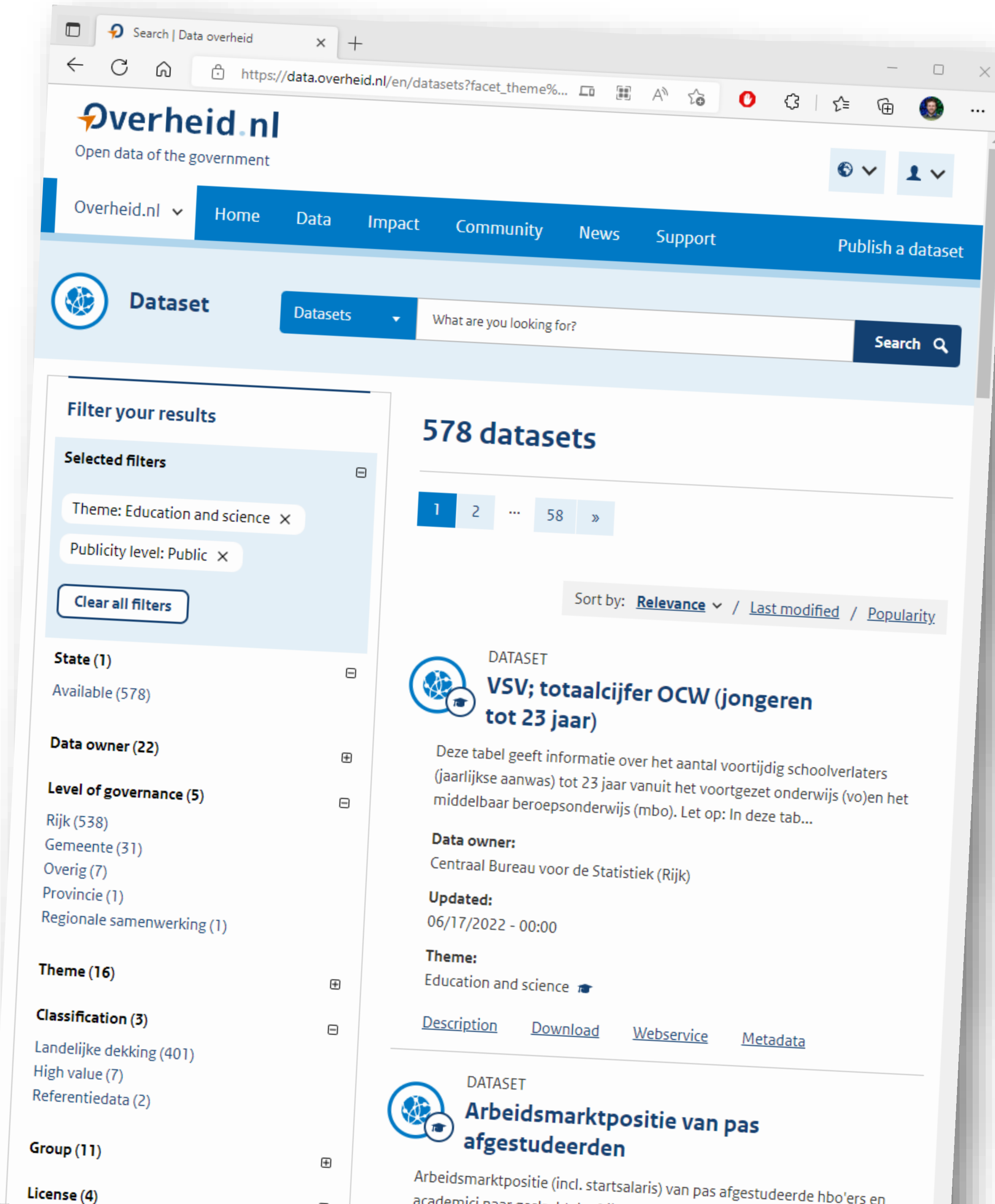


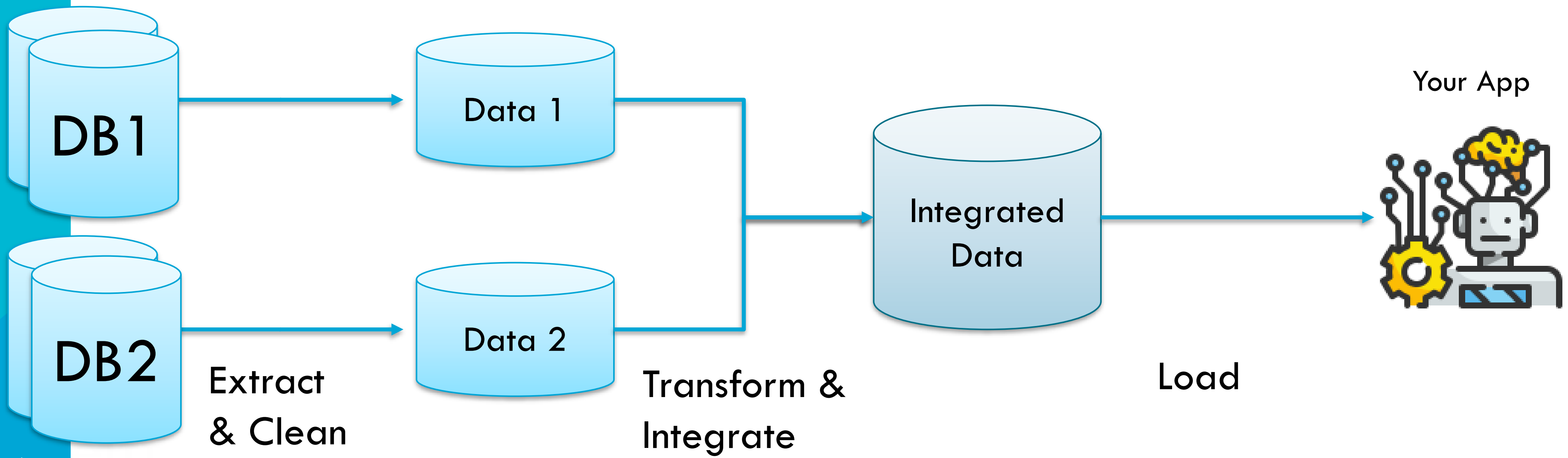


3. Your own Enterprise Database Management System

- “Make your own data”
- You might already own data yourself in your enterprise/app!
  - If you run a web shop, you already have data on customers, items, and purchases
  - If you run a community site, you already have data on user content and user interactions
- How extract that data from your database systems?
- Also, is this really your data? Are you allowed to use it?
  - See General Data Protection Regulation (GDPR) or European Data Act

3. Your own Enterprise Database Management System

- We often talk a about Data Pipelines in this context
- Or ETL: Extract, Transform, Load



4. Data Farms / Crowdsourcing

- Especially, labels are very hard to come by
  - Many labeling tasks require a real (intelligent?) person
  - e.g., detect complex objects in images, detect hate speech, detect irony, annotate emotions, etc.
- Coordinate many people to make labels /data for you
  - …recruiting paid or unpaid volunteers > Crowdsourcing
  - …as a centralized large-scale business > Data Farm

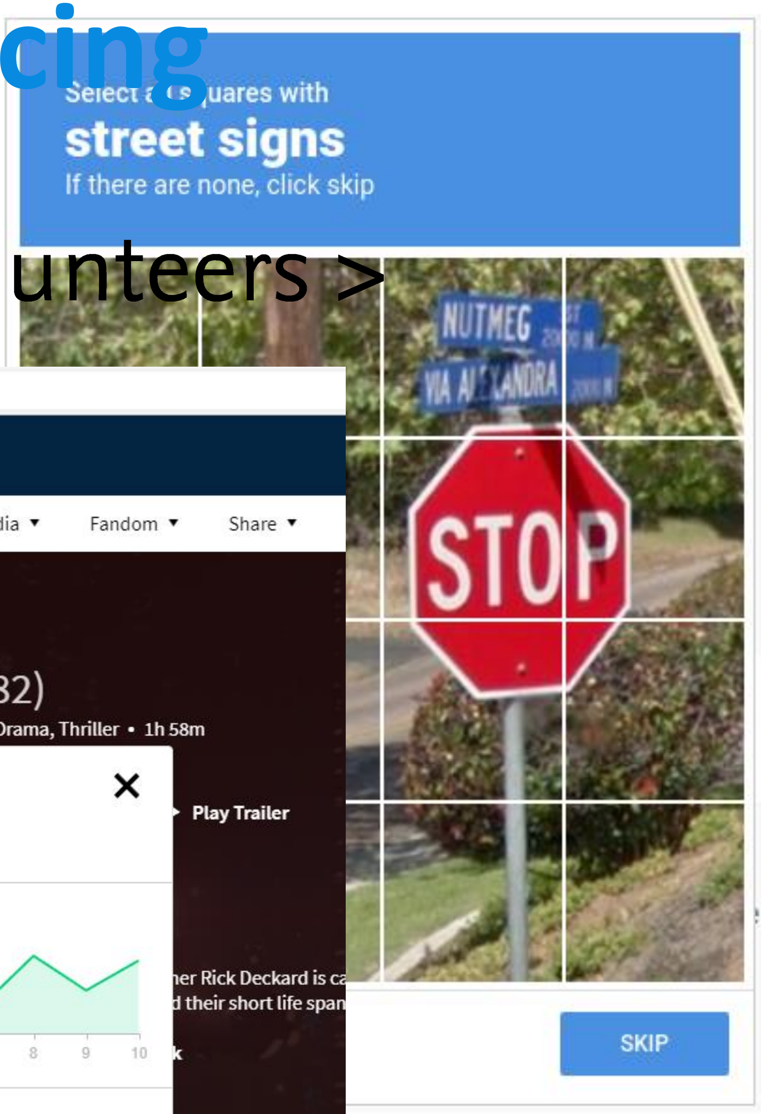
4. Data Farms / Crowdsourc

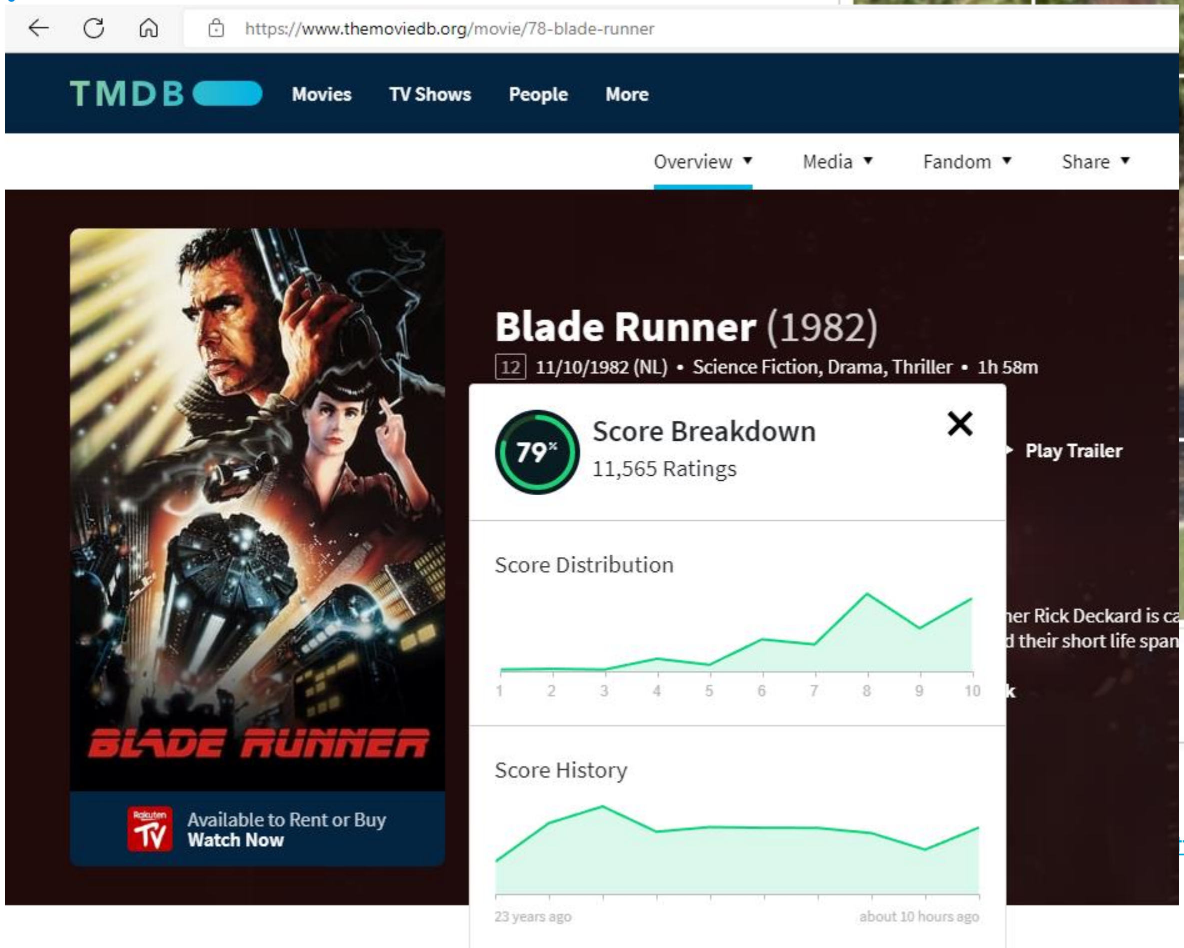
- …"recruiting" paid or unpaid vol
  Crowdsourcing

Piggybacking
From Google reCAPTCHA: https://www.google.com/recaptcha/

4. Data Farms / Crowdsourcing

- …"recruiting" paid or unpaid volunteers > Crowdsourci





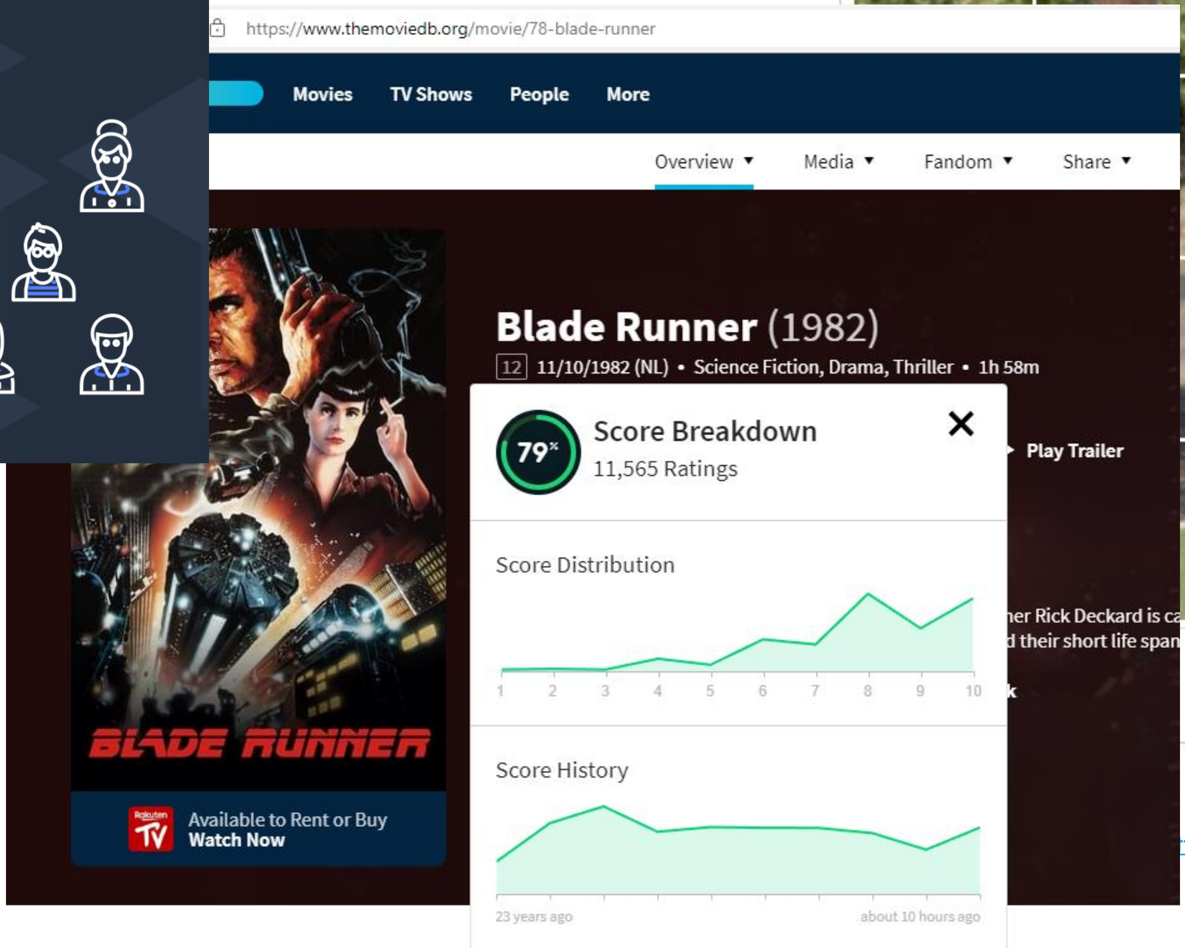
tps://www.google.com/recaptcha/

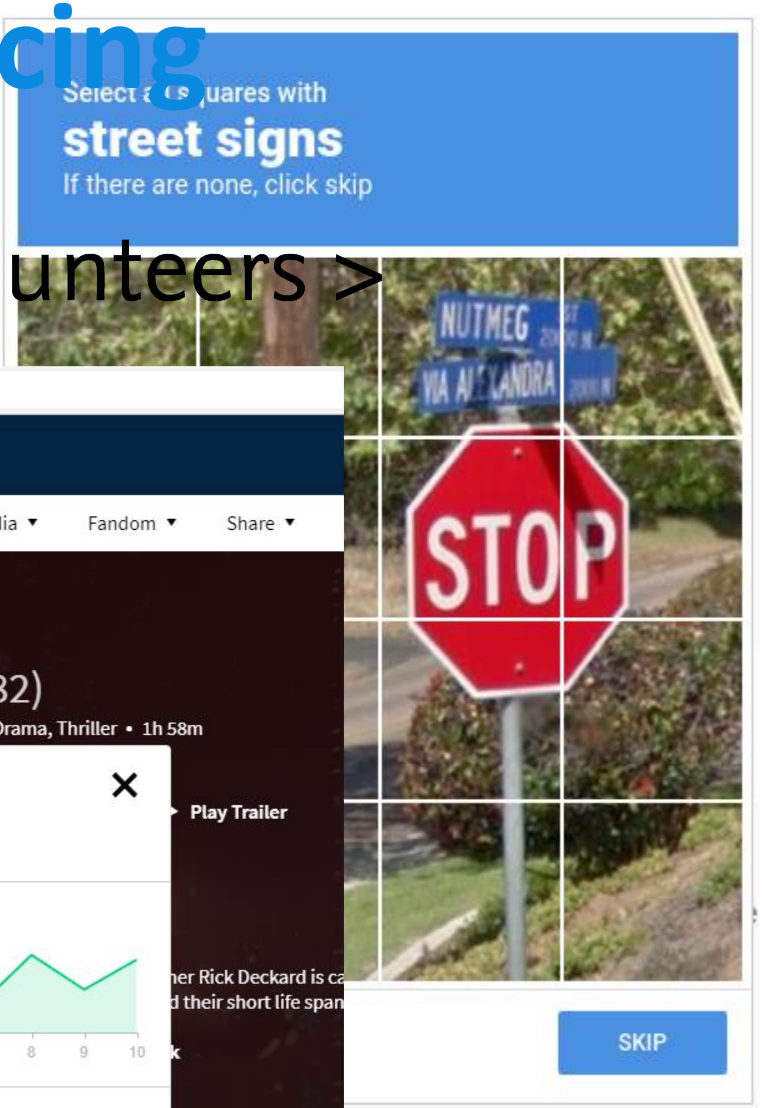
Community Contributions from TheMovieDB: https://www.themoviedb.org/movie/78-blade-runner

4. Data Farms / Crowdsourcing

…id or unpaid volunteers >

Paid Microtask Crowdsourcing from Amazon Mechanical Turk: https://requester.mturk.com/





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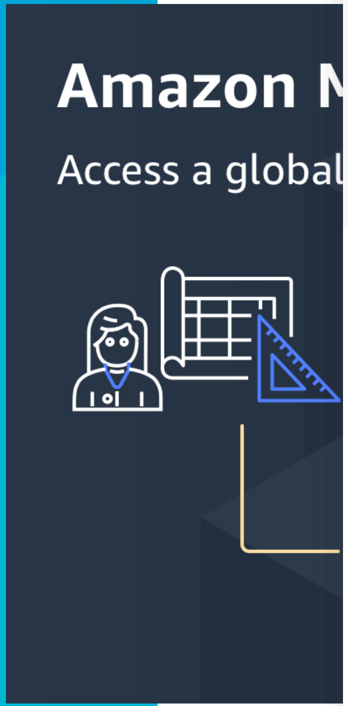
Community Contributions from TheMovieDB: https://www.themoviedb.org/movie/78-blade-runner

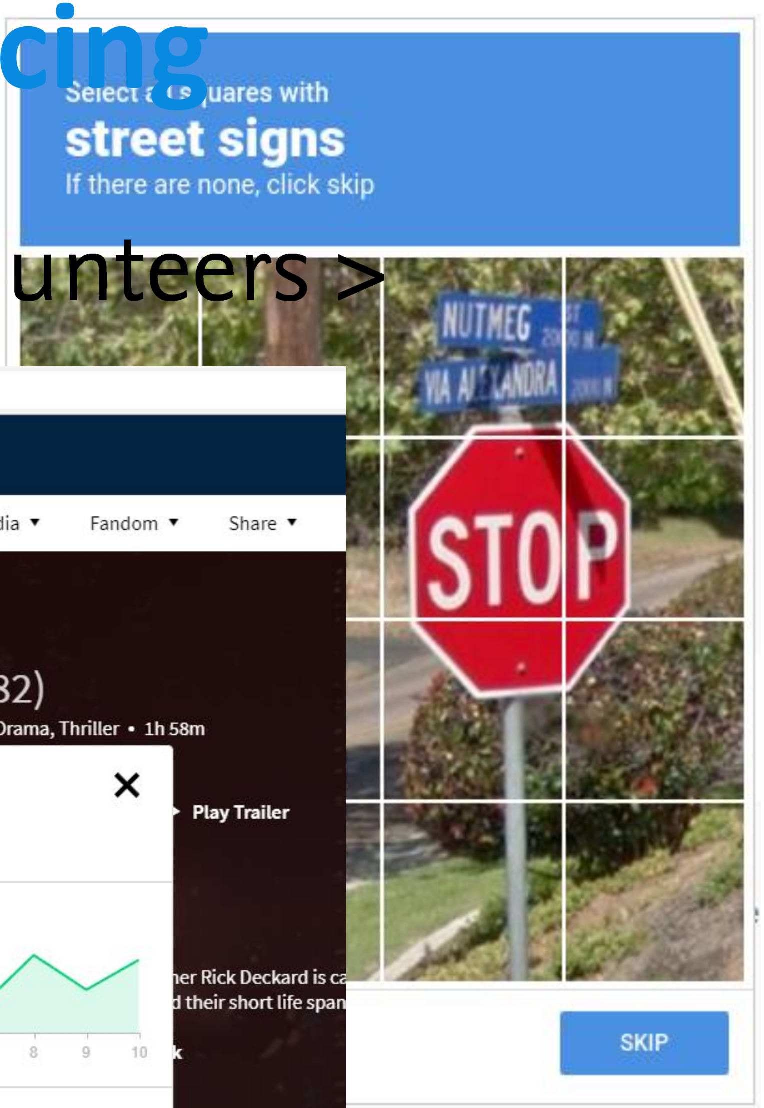
Citizen Science from Naturlais Leiden: https://www.naturalis.nl/en/about-us/collaborate/national-bee-count

4. Data Farms / Crowdsourcing

volunteers >



Paid Microtask Crowdsourcing from Amazon Mechanical Turk: http



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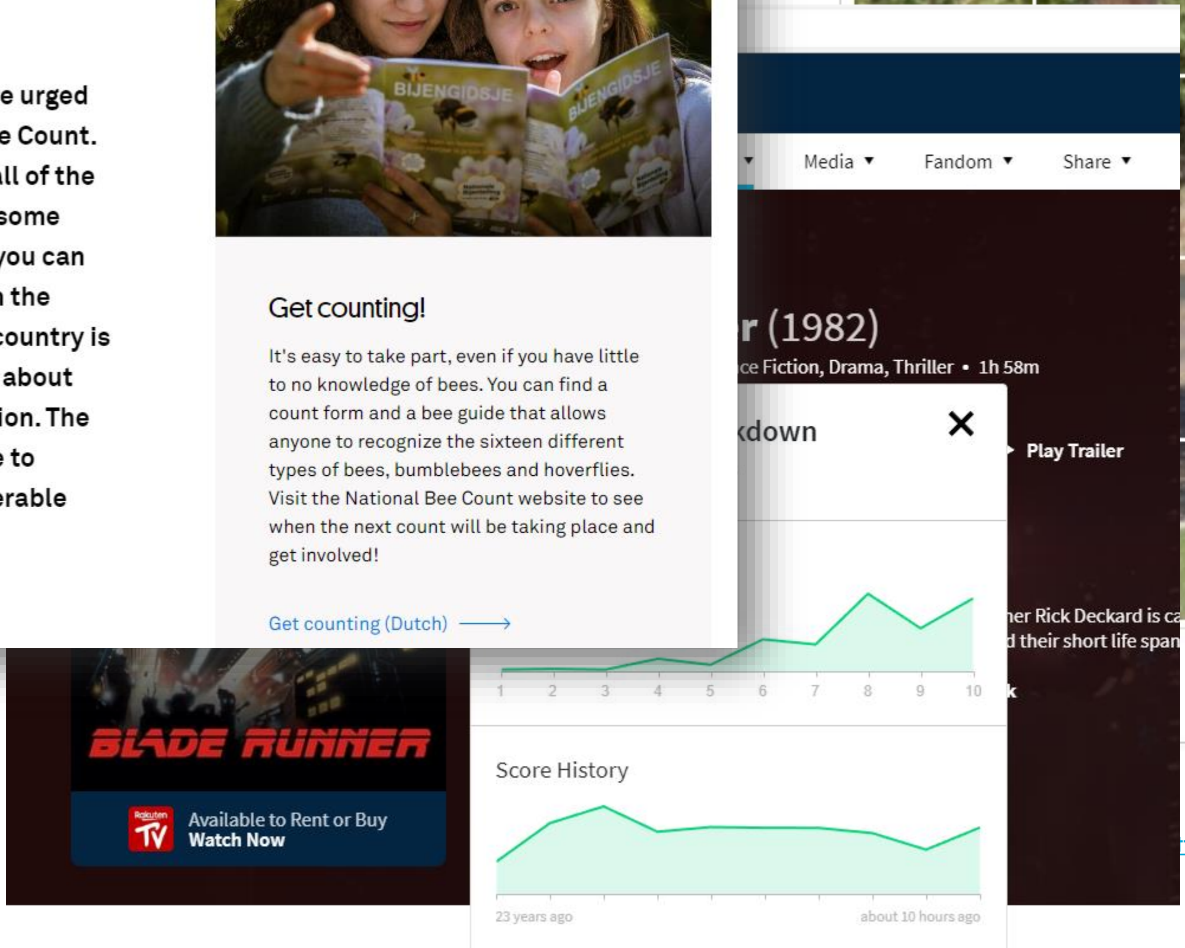


Community Contributions from TheMovieDB: https://www.themoviedb.org/movie/78-blade-runner

4. Data Farms / Crowdsourcing

from The Economist Jan 2020
https://www.economist.com/technology-quarterly/2020/01/02/chinas-success-at-ai-has-relied-on-good-data

Technology Quarterly
Jan 2nd 2020 edition ›

Data

China's success at AI has relied on good data

But cheap labour has also played an important part

CHINA IS THE land of face recognition. Cameras able to extract face prints from passers-by are common in the streets of large cities like Guangzhou and Shenzhen. Boxy vending machines at airports offer to let you pay for a cup of orange juice, robot-squeezed for perfect freshness, by scanning your face. From December 1st all people applying for an account with one of China's telecoms companies such as China Mobile must have their face scanned. Previous regulations required proof of identity, but the possession of users' face prints will let firms verify identities in real-time via smartphone cameras.

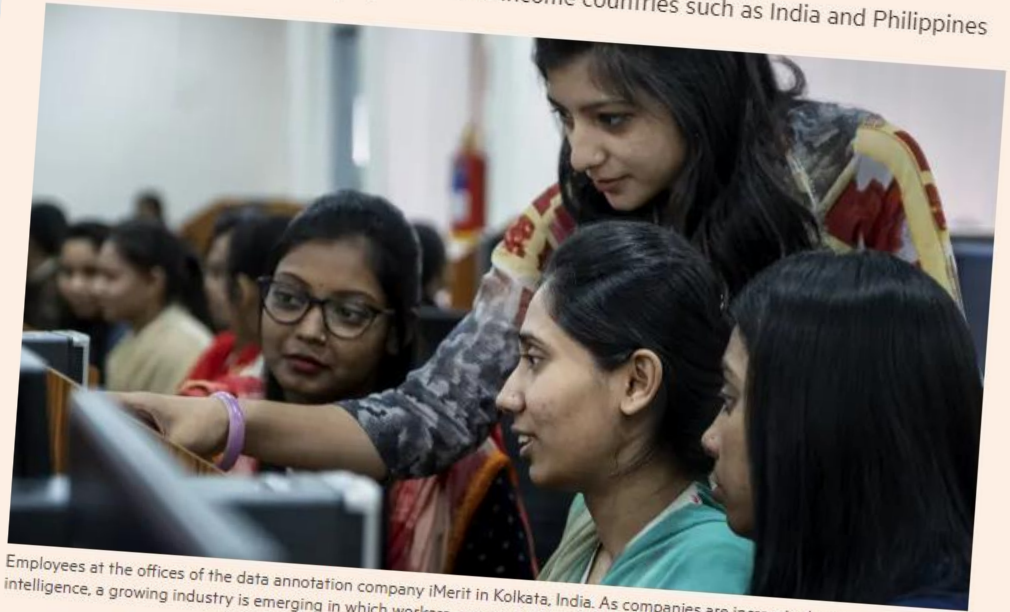
Considering the oppressive purposes to which this technology is being put—most notably in the Muslim-majority areas of north-west China—it would not be appropriate to call China's rapid adoption of it anything more than a technical success. The underappreciated fact that companies leaping ahead in the field are reliant on cleverly deployed cheap labour for their progress than on any

from The Financial Times July 2019
https://www.ft.com/content/56dde36c-aa40-11e9-984c-fac8325aaa04

Artificial intelligence + Add to myFT

AI's new workforce: the data-labelling industry spreads globally

Hundreds of thousands employed in lower-income countries such as India and Philippines



Employees at the offices of the data annotation company iMerit in Kolkata, India. As companies are increasingly embracing artificial intelligence, a growing industry is emerging in which workers are employed to 'train' algorithms to recognise various types of data

Madhumita Murgia JULY 24 2019

26

On the fringes of the Indian city of Kolkata, in the dusty, crowded neighbourhood of Metiabruz, 460 young women are working at the vanguard of artificial intelligence.

The women, mostly from the local Muslim community, are helping to train computer vision algorithms used in autonomous vehicles and augmented reality systems, for the likes of Amazon, Microsoft, eBay and T…

from The New York Times Nov 2018: https://www.nytimes.com/2018/11/25/business/china-artificial-intelligence-labeling.html

4. Data l

Technology Quarterly

Jan 2nd 2020 edition ›

Data

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BUSINESS

The New York Times

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THE NEW NEW WORLD

How Cheap Labor Drives China's A.I. Ambitions

Give this article

17

Workers at the headquarters of Ruijin Technology Company in Jiaxian, in central China's Henan Province. They identify objects in images to help artificial intelligence make sense of the world. Yan Cong for The New York Times

By Li Yuan

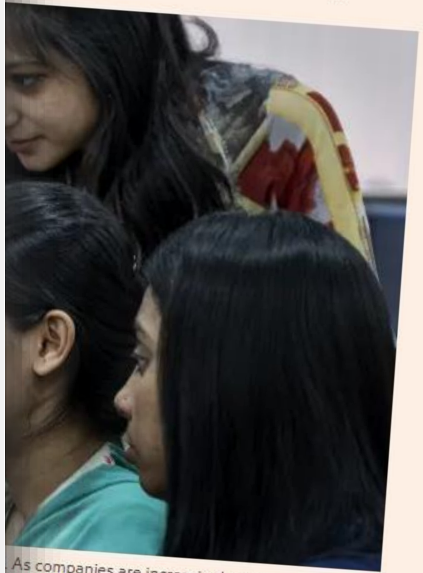
Nov. 25, 2018

阅读简体中文版 · 閱讀繁體中文版

Some of the most critical work in advancing China's technology goals takes place in a former cement factory in the middle of the

ta-labelling industry

ountries such as India and Philippines



As companies are increasingly embracing artificial
ain' algorithms to recognise various types of data

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dusty, crowded
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from The Verge Feb 2019

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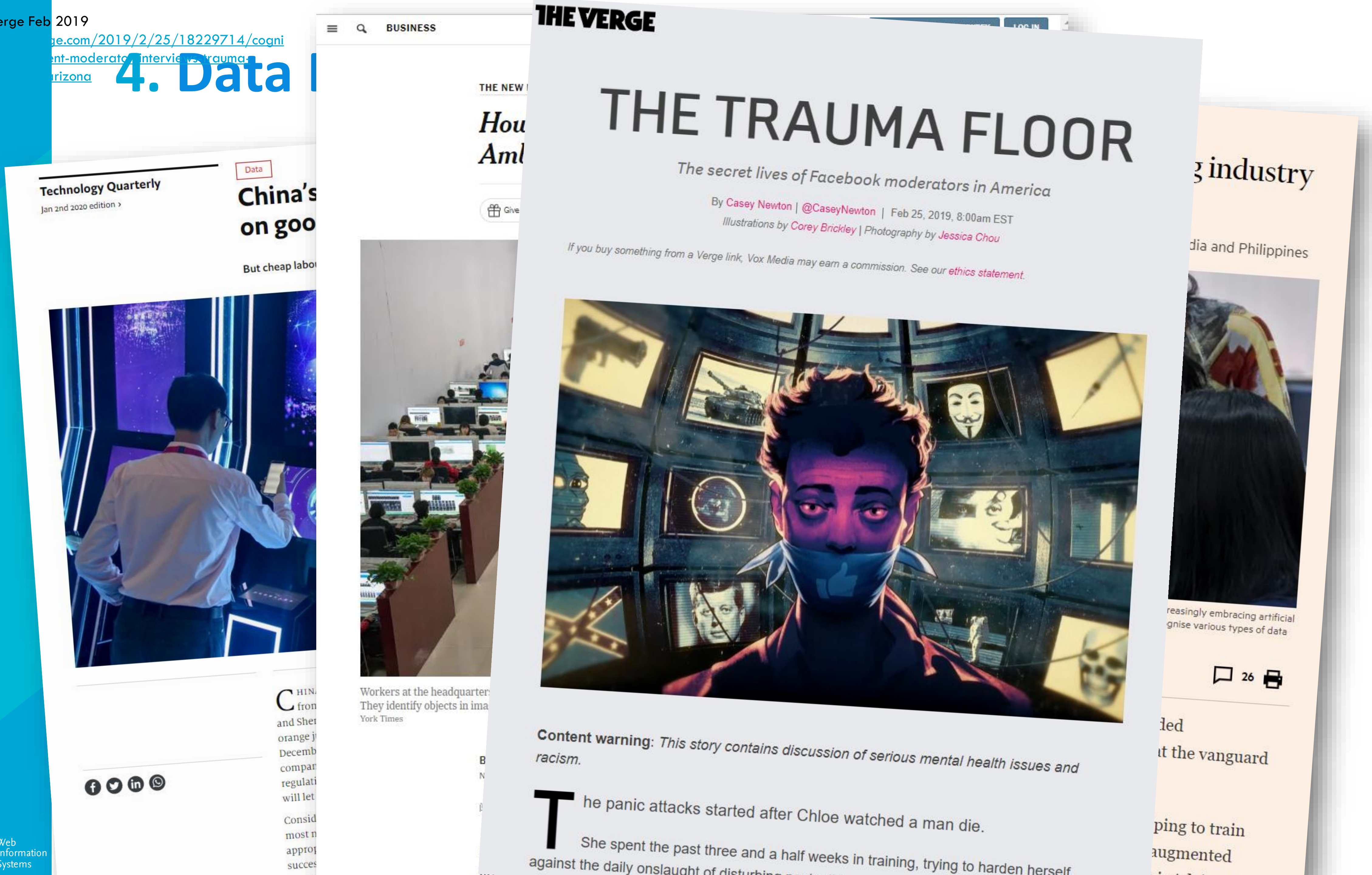
4. Data l



But what if you never had the data to start with?

- From the Economist, and Cognilytica:
  - 3rd party data preparation market was 1.5bn USD in 2019, expected 3.5bn USB in 2024
  - 3rd party labeling 1.7bn USD 2019, 4.1bn USD in 2024

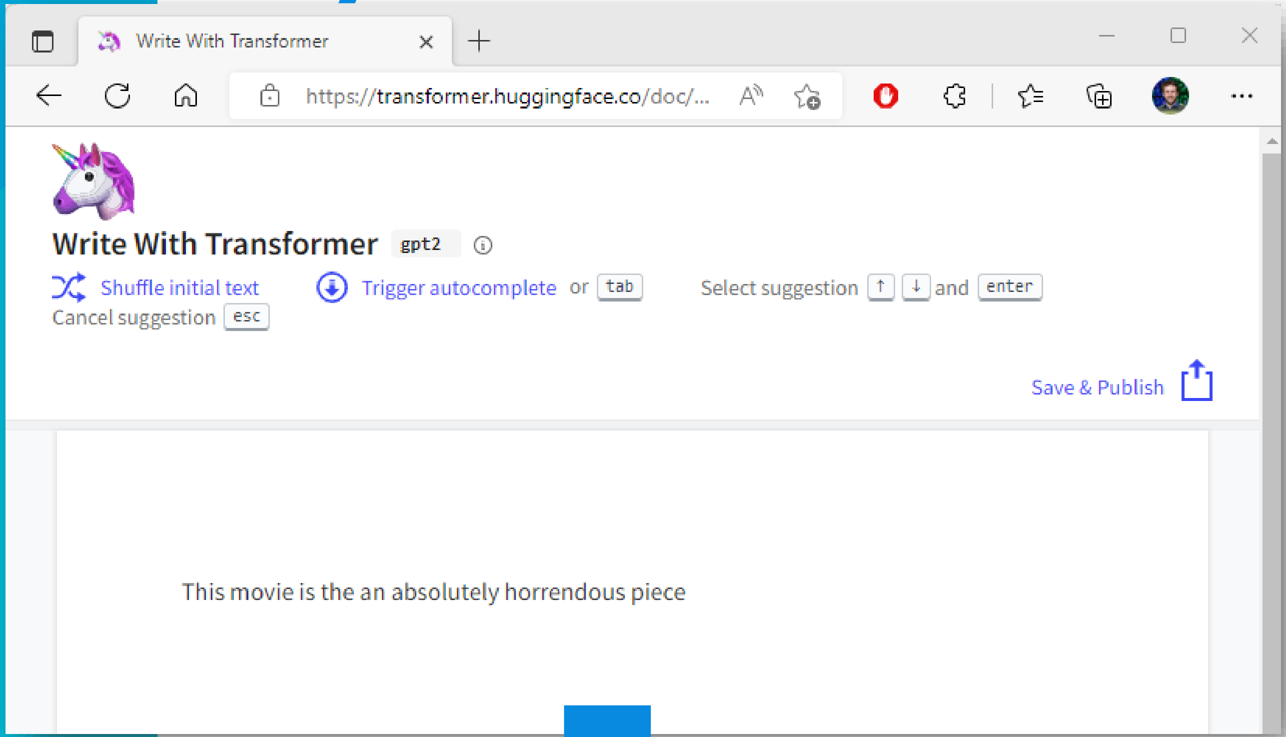
https://www.economist.com/technology-quarterly/2020/06/11/for-ai-data-are-harder-to-come-by-than-you-think

5. Artificial / Synthetic Data

- Hot academic topic!
- Core Idea: Train an AI to learn a representation of your data, and then use that to generate more like it
  - For example, using Variational Autoencoders
- Such data is not "real", and thus "lower quality"
  - …but you can make a lot of it, and often volume beats quality!
- Works well for augmenting real datasets
  - e.g., balance under-represented classes

Synthetic Text

from Hugginface / GPT2:
https://transformer.huggingface.co/doc/gpt2-large
https://openai.com/blog/tags/gpt-2/

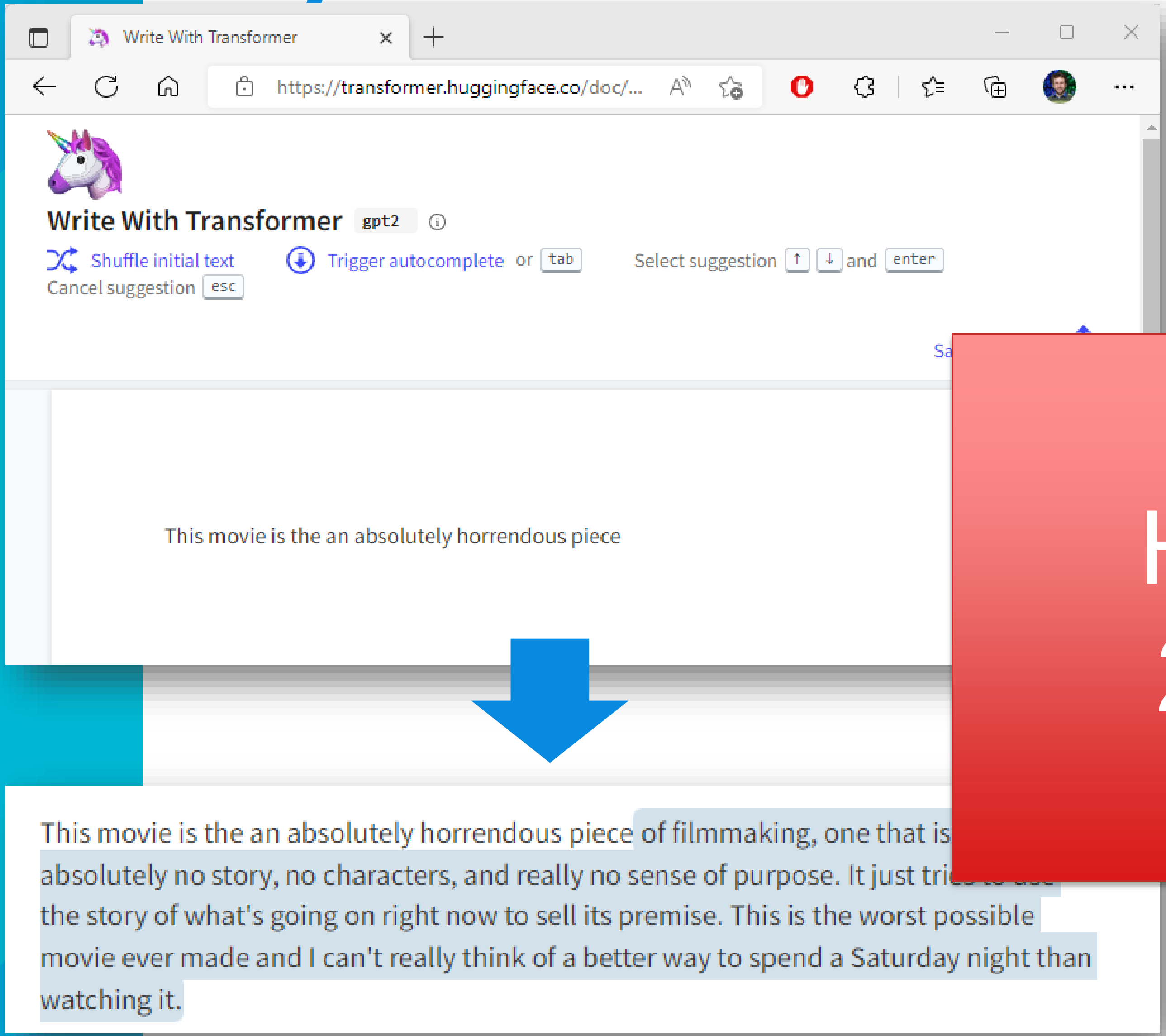


This movie is the an absolutely horrendous piece of filmmaking, one that is filled with absolutely no story, no characters, and really no sense of purpose. It just tries to use the story of what's going on right now to sell its premise. This is the worst possible movie ever made and I can't really think of a better way to spend a Saturday night than watching it.

???!

Synthetic Text

from Hugginface / GPT2:
https://transformer.huggingface.co/doc/gpt2-large
https://openai.com/blog/tags/gpt-2/

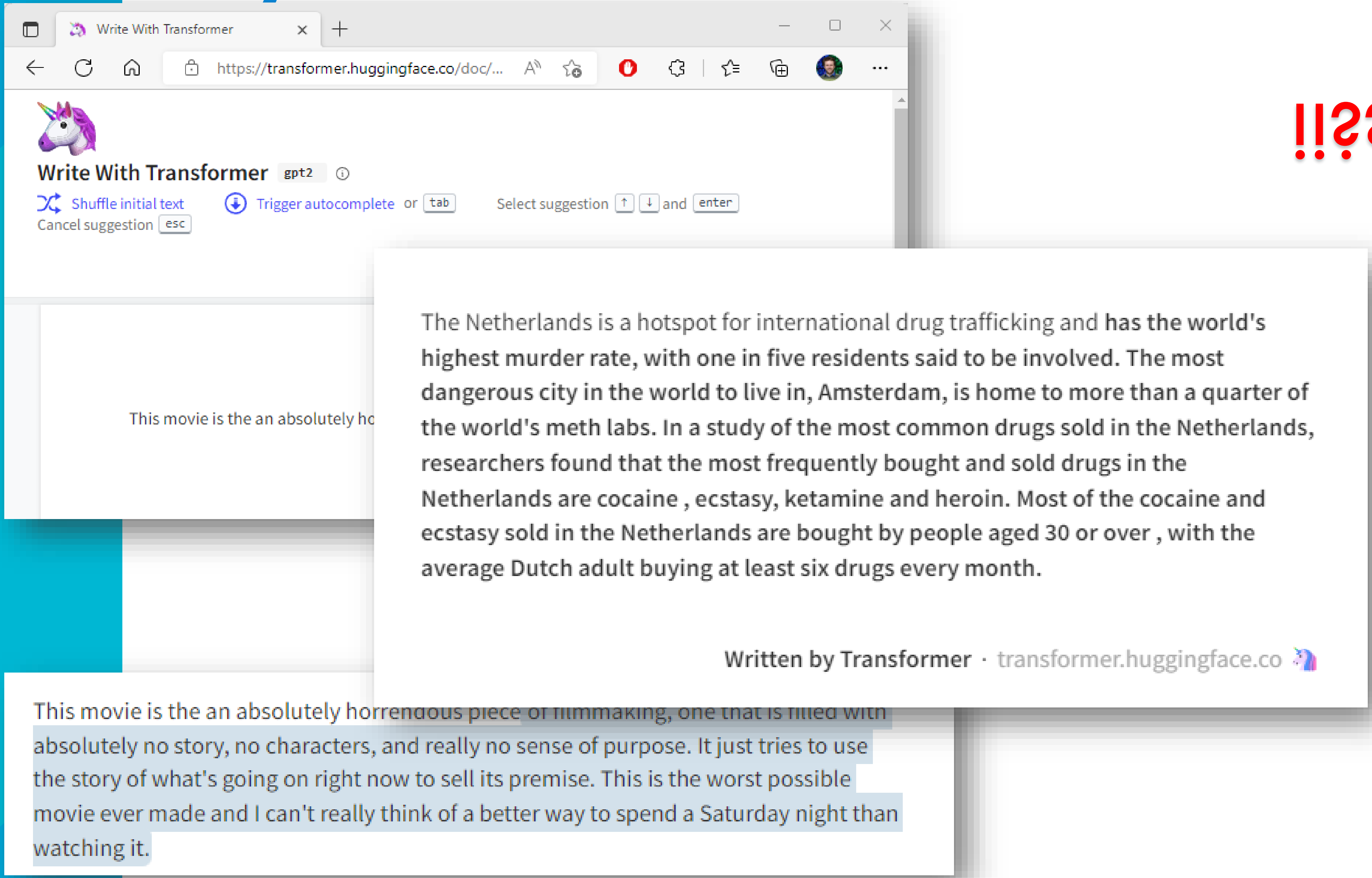


HAHA! Slides from 2022…time flies!!!

???!

Synthetic Text

from Hugginface / GPT2:
https://transformer.huggingface.co/doc/gpt2-large
https://openai.com/blog/tags/gpt-2/

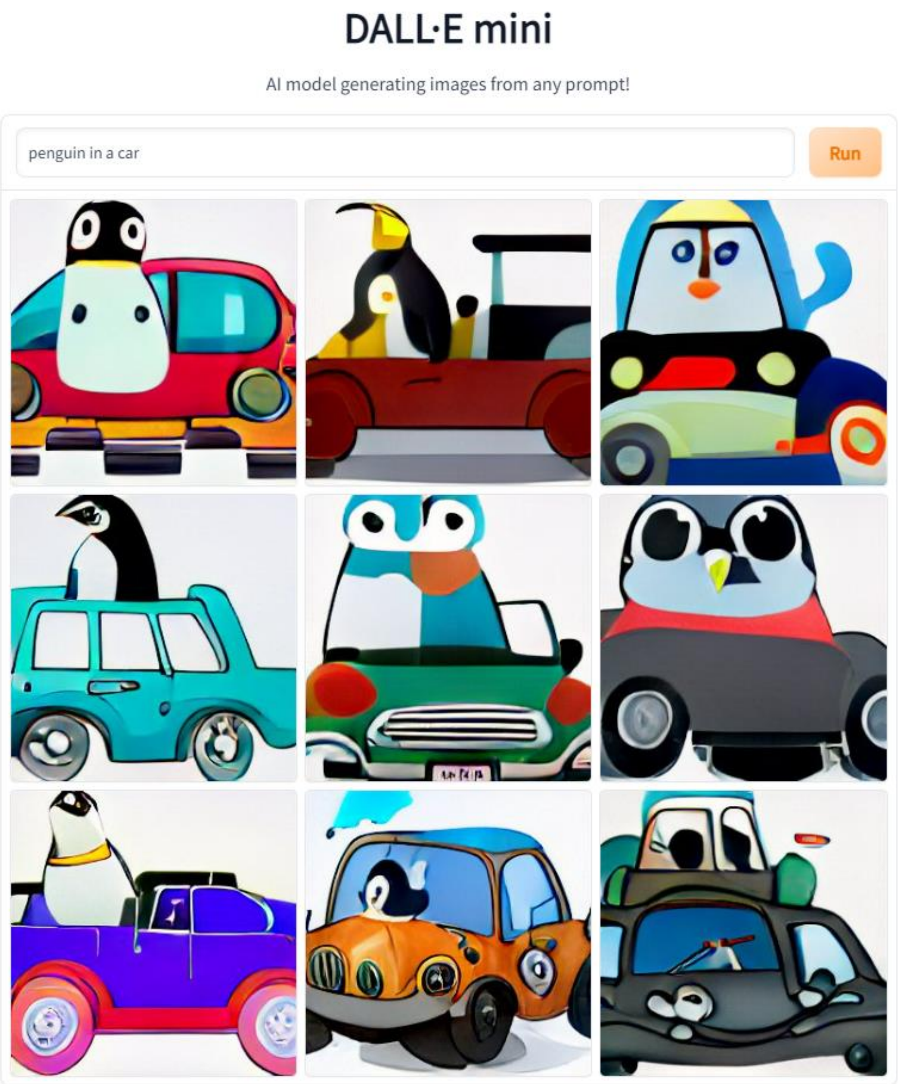


!!??!!!

Synthetic Images

from Hugginface / OpenAI:
DALL·E mini - by craiyon.com (huggingface.co)
https://openai.com/dall-e-2/

"Penguin in a car"



"Bicycle made of cheese"

- 2024 ☺

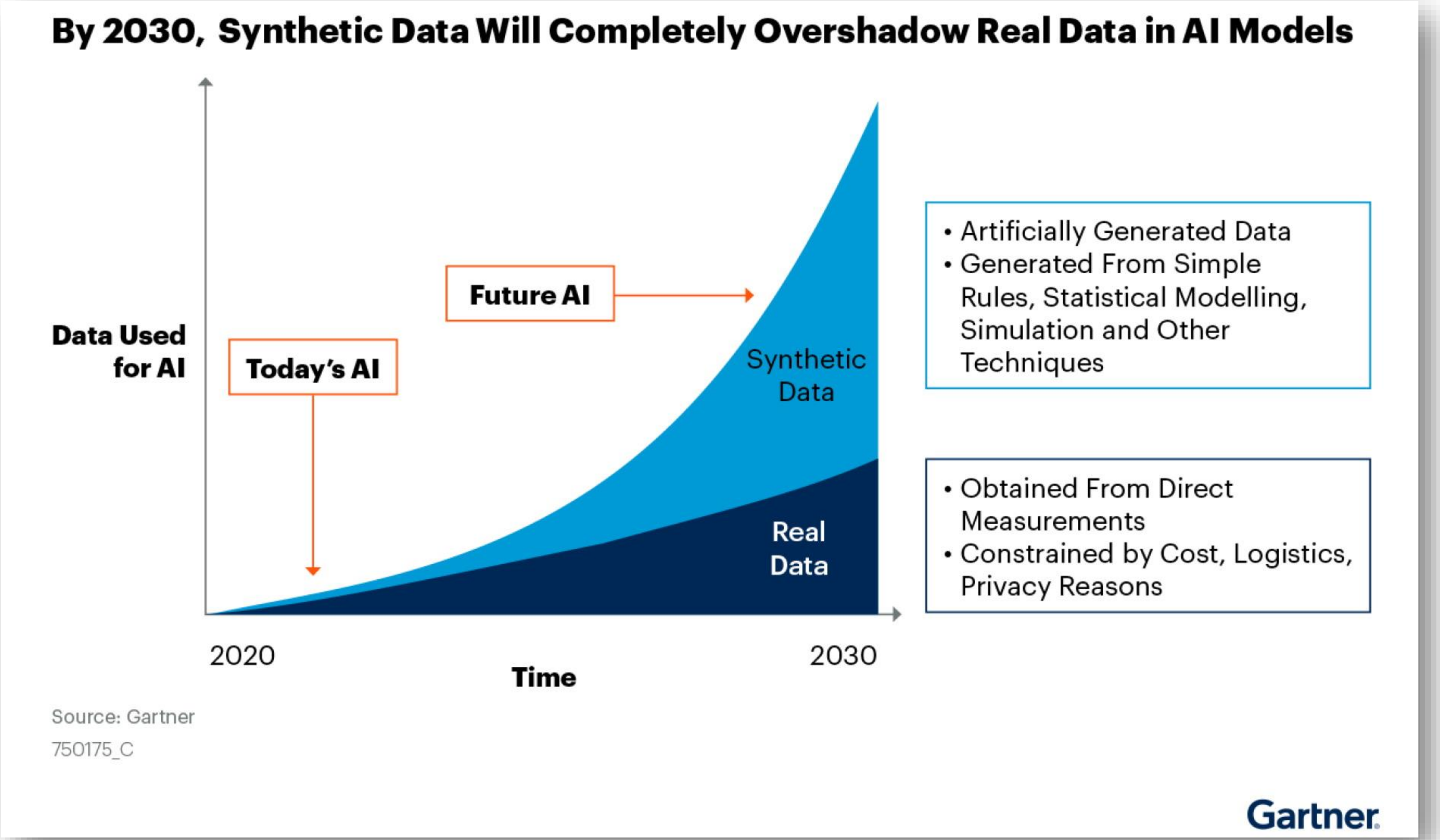
Synthetic Images

from: Ramesh, Aditya, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. "Zero-shot text-to-image generation." In *International Conference on Machine Learning*, pp. 8821-8831. PMLR, 2021.
Also: https://cv-tricks.com/how-to/dall-e-text-to-image-generation/

"A road sign with an image of a blue strawberry"

Synthetic Images 2024 Edition

A road next to Institut für Informatik in Zürich / Oerlikon clearly showing a traffic sign with a blue strawberry on it. the sign has a circle shape, and the strawberry is heavily stylized and symbolized.



From:
Synthetic data will become the main form of data used in AI. Source: Gartner, "Maverick Research: Forget About Your Real Data – Synthetic Data Is the Future of AI," Leinar Ramos, Jitendra Subramanyam, 24 June 2021

6. “Steal it”…or obtain in other dubious ways

LAUREN GOODE BUSINESS SEP 25, 2023 6:47 PM

Getty Images Plunges Into the Generative AI Pool

Stock-photo giant Getty Images has partnered with Nvidia to build an image generator. Just like with other tools of its ilk, questions remain about who should get credit for the pictures it dreams up.

ILLUSTRATION: ANDRIY ONUFRIYENKO/GETTY IMAGES

https://www.wired.com/story/getty-images-generative-ai-photo-tool/

6. “Steal it”…or obtain in other dubious ways

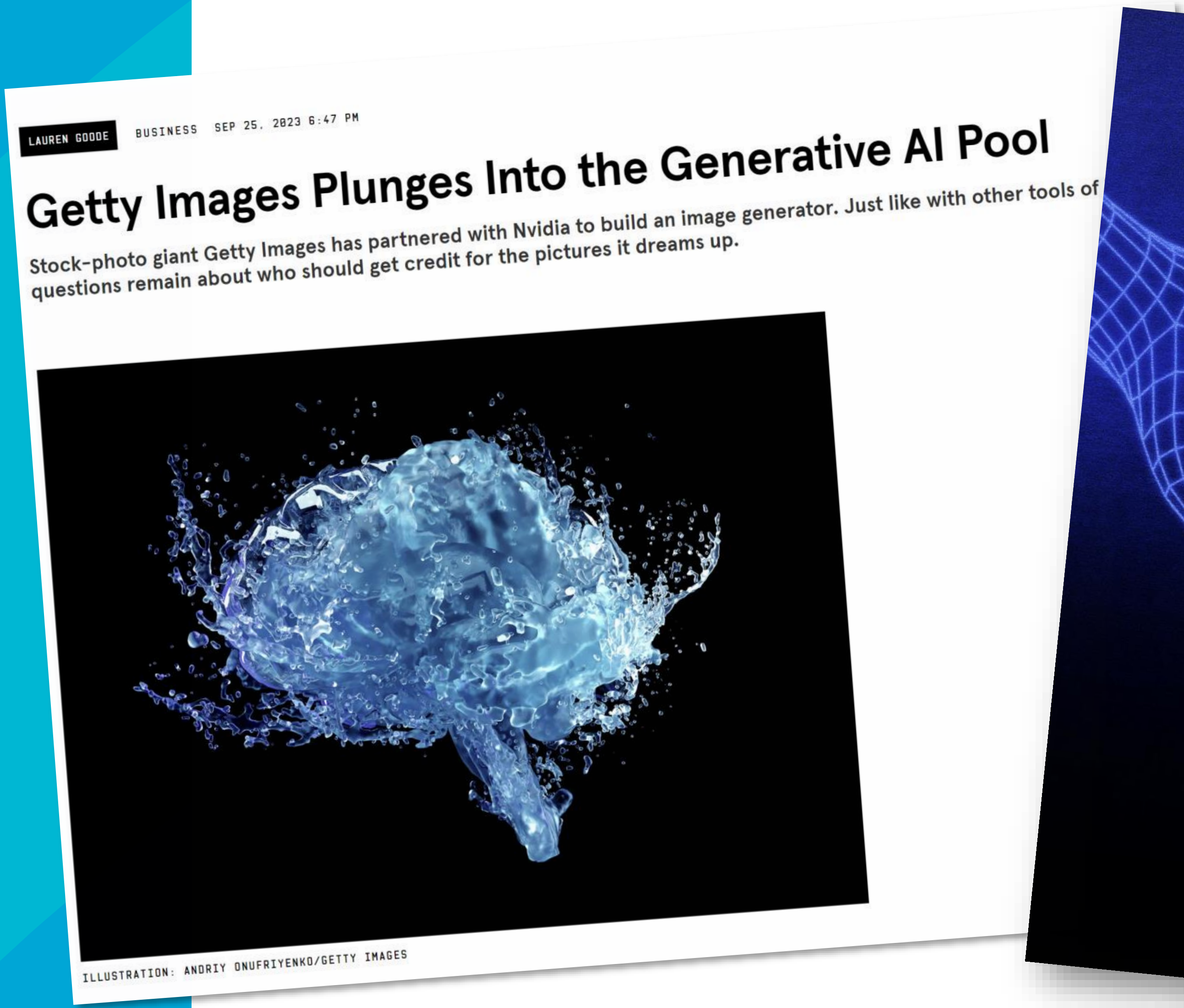

LAUREN GOODE BUSINESS SEP 25, 2023 6:47 PM

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ILLUSTRATION: ANDRIY ONUFRIYENKO/GETTY IMAGES

https://www.newyorker.com/culture/infinite-scroll/is-ai-art-stealing-from-artists

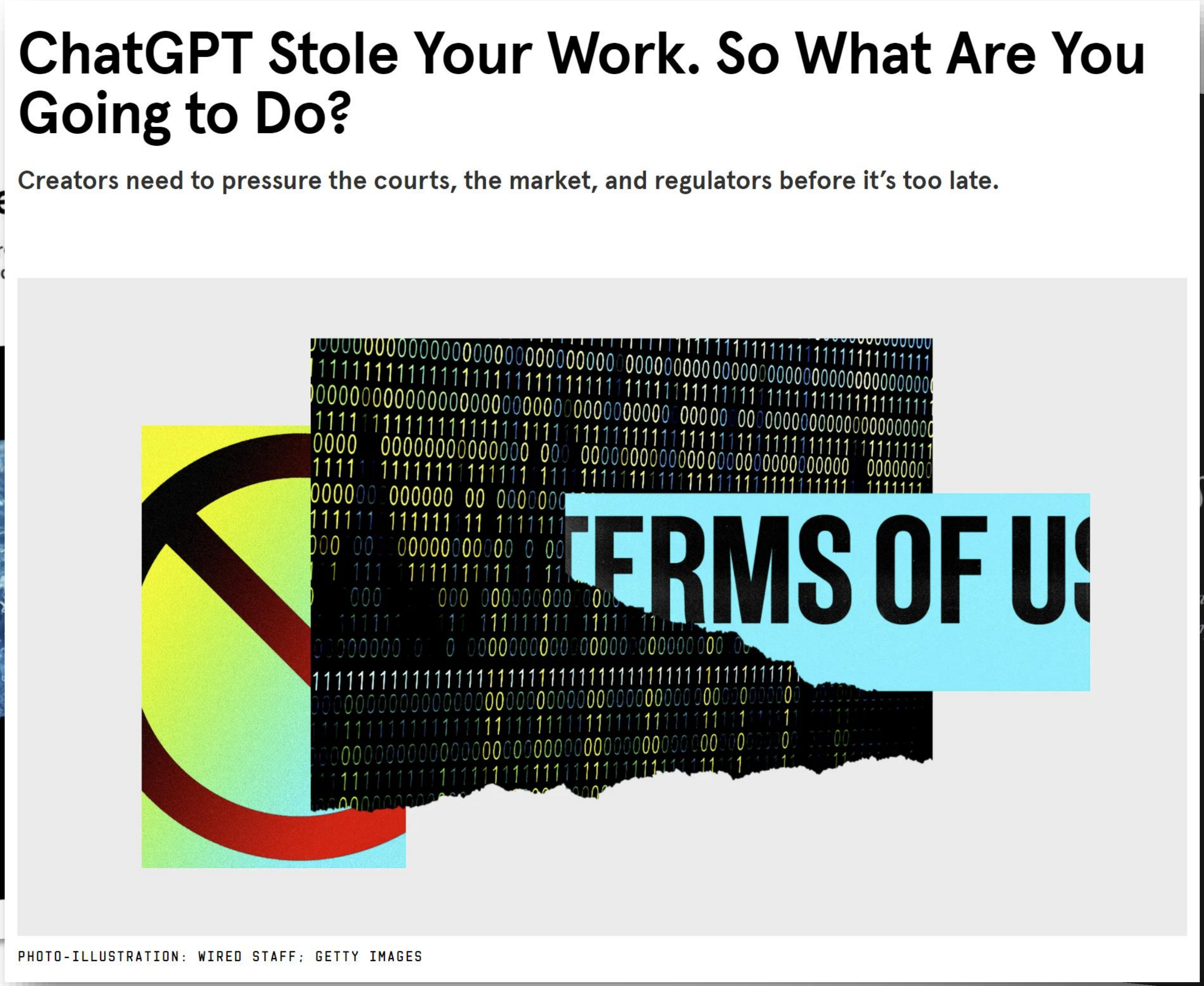
6. "Steal it"…or obtain in other dubious ways


LAUREN GOODE BUSINESS SEP 25, 2023 6:47 PM

Getty Images Plunge

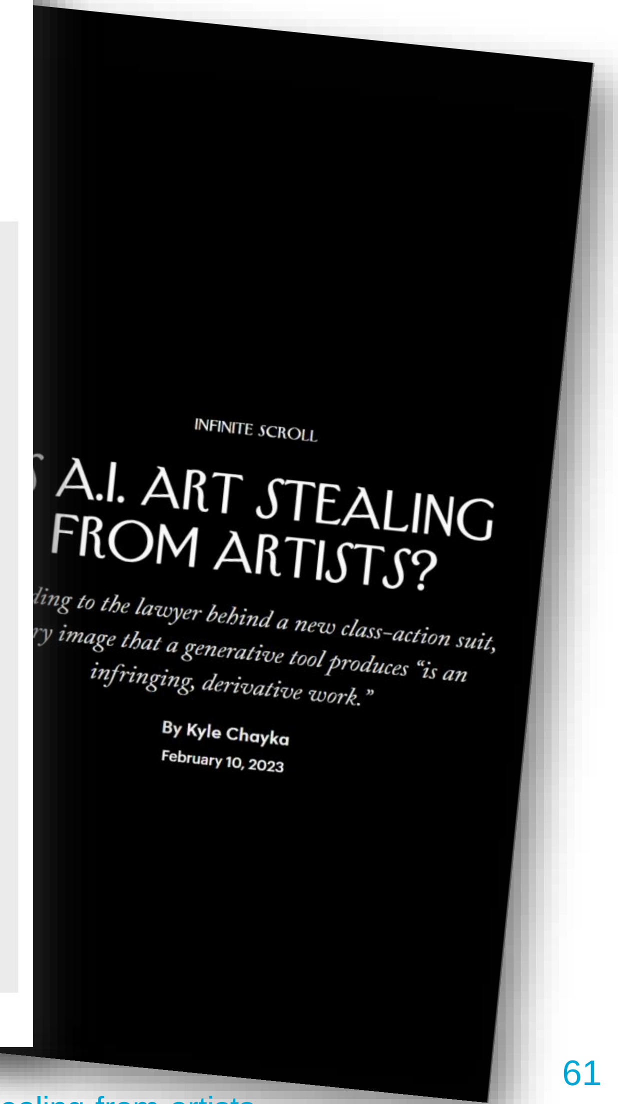
Stock-photo giant Getty Images has partner
questions remain about who should get cred

ILLUSTRATION: ANDRIY ONUFRIYENKO/GETTY IMAGES


ChatGPT Stole Your Work. So What Are You Going to Do?

Creators need to pressure the courts, the market, and regulators before it's too late.

PHOTO-ILLUSTRATION: WIRED STAFF; GETTY IMAGES


INFINITE SCROLL

A.I. ART STEALING FROM ARTISTS?

ding to the lawyer behind a new class-action suit, ry image that a generative tool produces "is an infringing, derivative work."

By Kyle Chayka

February 10, 2023

https://www.newyorChatGPT Stole Your Work. So What Are You Going to Do? | WIREDker.com/culture/infinite-scroll/is-ai-art-stealing-from-artists

6. “Steal it”

ENTERTAINMENT > MOVIES

https://people.com/sag-claims-studios-want-to-scan-background-actors-use-likenesses-for-free-7561285

Hollywood Studios Want to Scan Background Actors for One Day's Pay, Use Likenesses for 'Rest of Eternity,' SAG Claims

In a statement amid the strike decision, the AMPTP said that their offer included "a groundbreaking AI proposal that protects actors' digital likenesses"

By **Jen Juneau** | Published on July 14, 2023 03:26PM EDT

PHOTO: JOHN SALANGSANG/SHUTTERSTOCK

6. "Steal it"…or obtain in other dubious ways

- We don't really know yet what to do here
  - Legal aspects?
    - What is allowed and forbidden?
    - How do we deal with this internationally???
  - Ethical aspects?
    - Just because it's legal doesn't mean that it's right

Is my AI system a sexist Nazi?

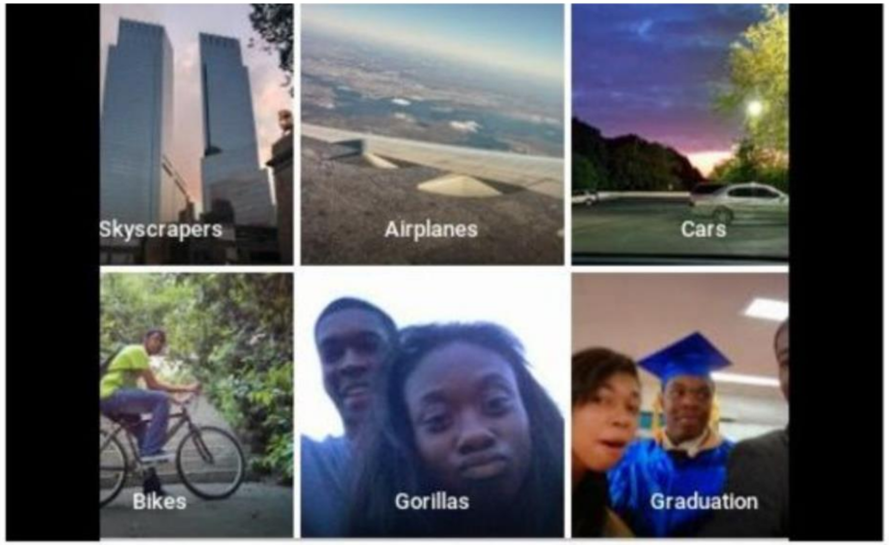
…or did it just feed on bad data?

Bias

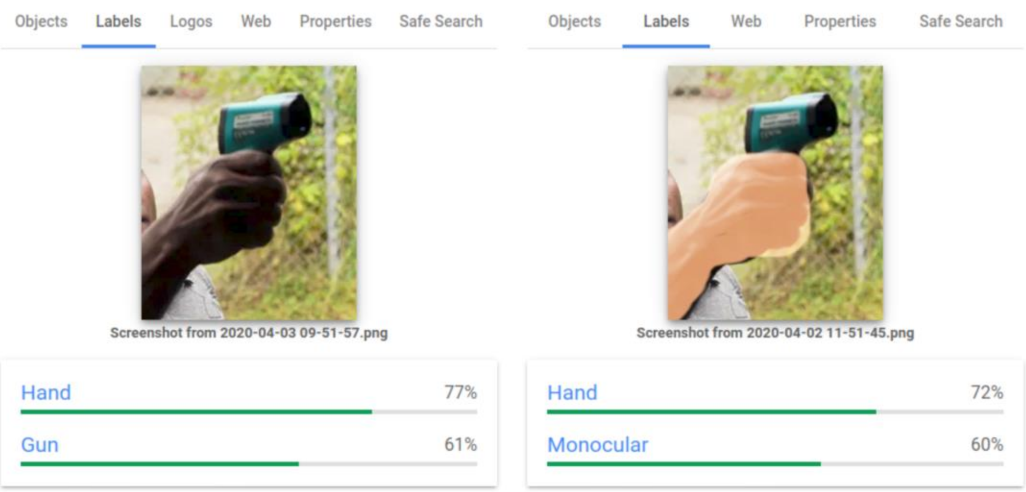
Google apologises for Photos app's racist blunder

1 July 2015

Share



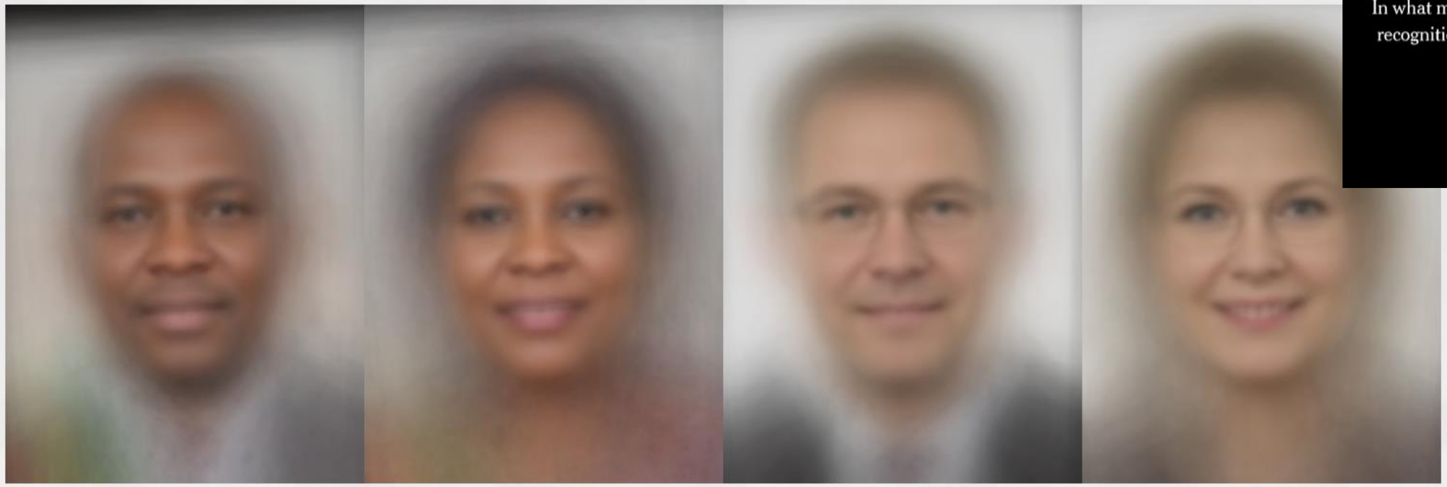
diri noir avec banan @jackyalcine · Jun 29
Google Photos, y'all [redacted] My friend's not a gorilla.

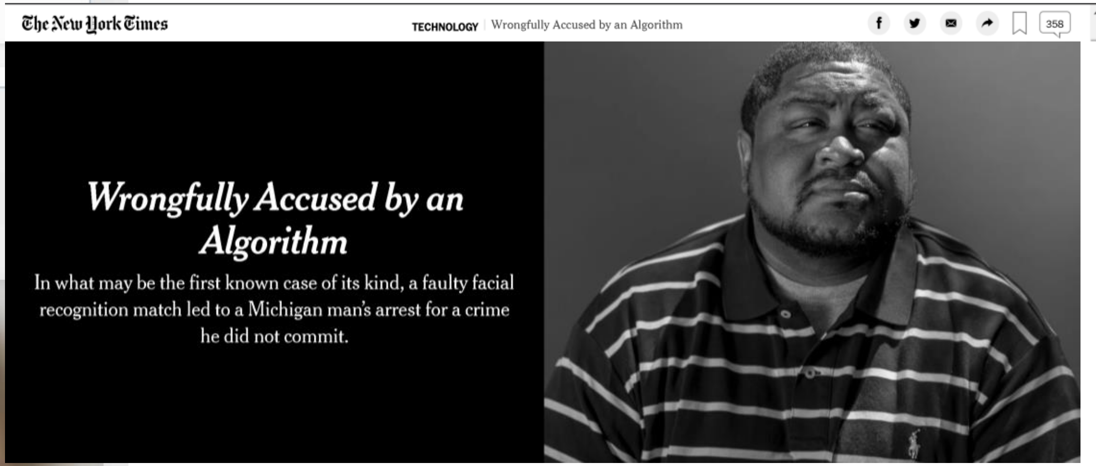


On 3 April, Google Vision Cloud produced starkly different labels after an overlay was added.

Large training datasets of possibly low diversity
Deep learning model learned wrong prediction features

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%

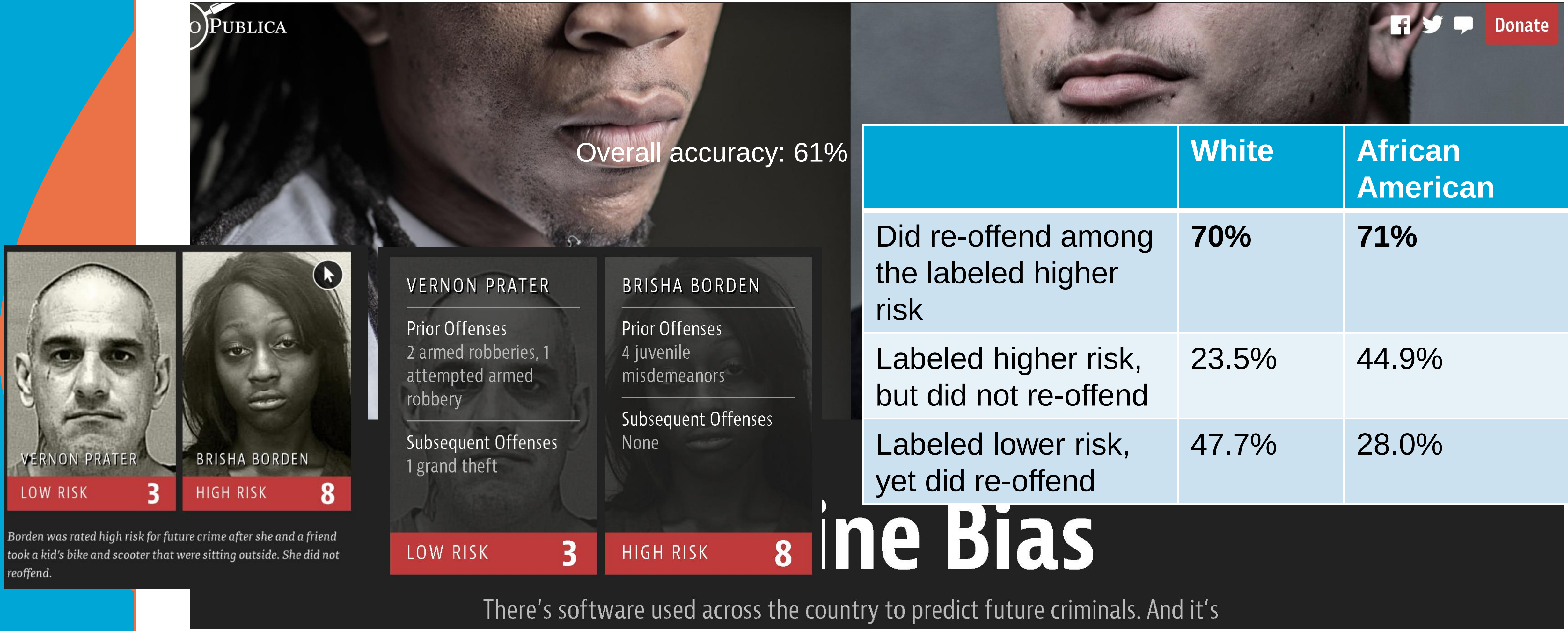




Unbalance in training data for different populations
→ Lack of requirements engineering

Attributes:
- defendant's demographic information (e.g., gender, race),
- criminal history (e.g., number of prior offenses)
- administrative information about the case (e.g., case number, arrest date, zipcode)

Improper modeling of domain knowledge and requirements:
- Use of irrelevant features
- Biased historical data



Overall accuracy: 61%

	White	**African American**
Did re-offend among the labeled higher risk	**70%**	**71%**
Labeled higher risk, but did not re-offend	23.5%	44.9%
Labeled lower risk, yet did re-offend	47.7%	28.0%

TU Delft WIS Web Information Systems

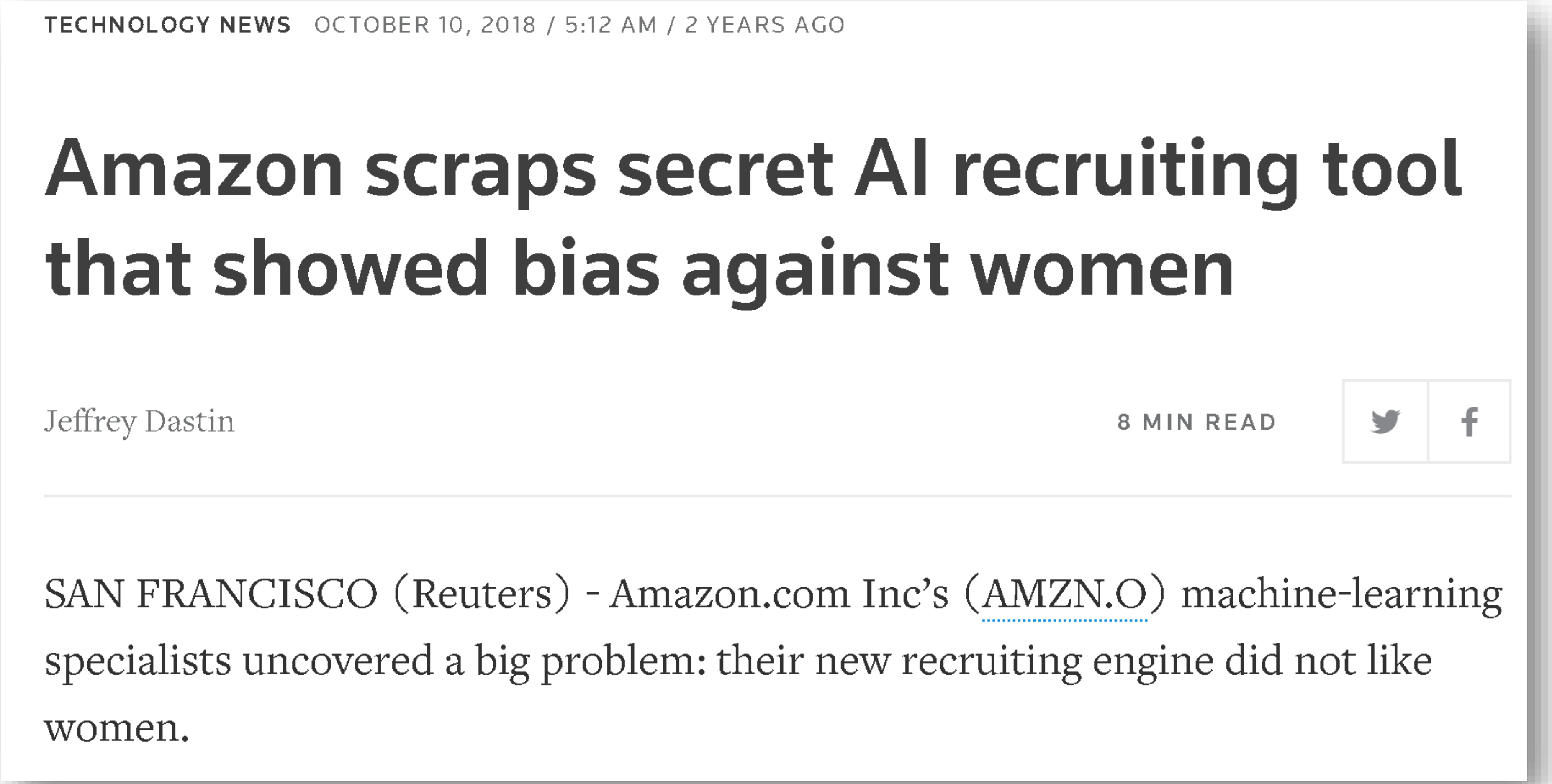
67

The New York Times

Apple Card Investigated After Gender Discrimination Complaints

A prominent software developer said on Twitter that the credit card was “sexist” against women applying for credit.

- Lack of proper use-case analysis
    - Not up-to-date dataset
    - Incorrect feature values
        - e.g. loans from a married couple registered only under men
        - → Data bias


TECHNOLOGY NEWS OCTOBER 10, 2018 / 5:12 AM / 2 YEARS AGO

Amazon scraps secret AI recruiting tool that showed bias against women

Jeffrey Dastin 8 MIN READ

SAN FRANCISCO (Reuters) - Amazon.com Inc's (AMZN.O) machine-learning specialists uncovered a big problem: their new recruiting engine did not like women.

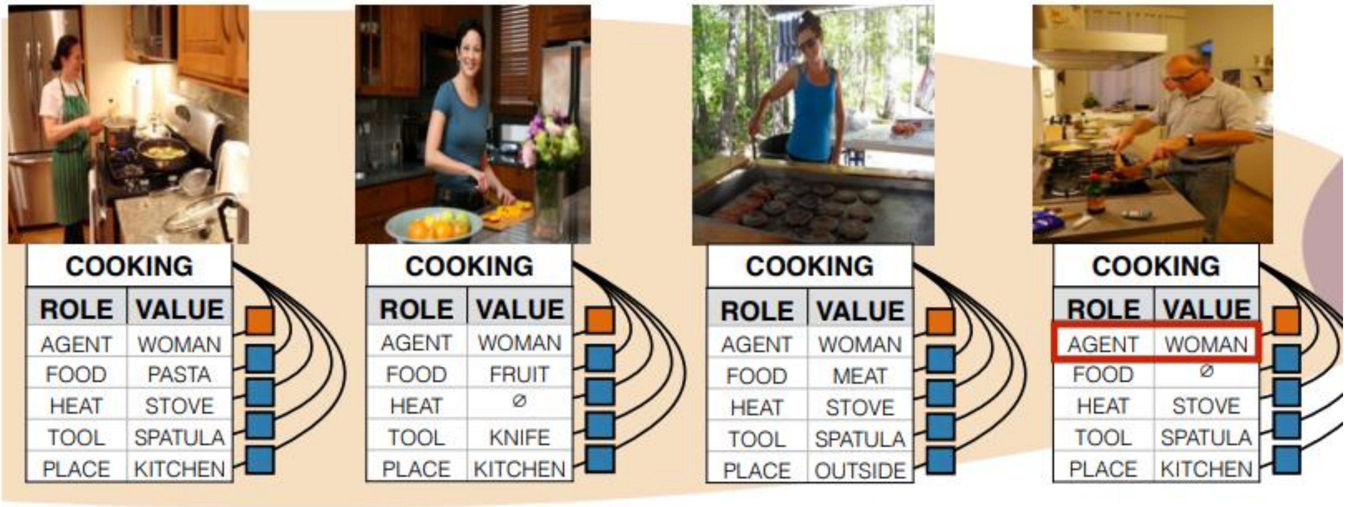
- Biased historical data:
  - Lack of data on the non-dominant group
  - Using an "objective" machine learning algorithm is not enough!
- Existence of proxy information for gender

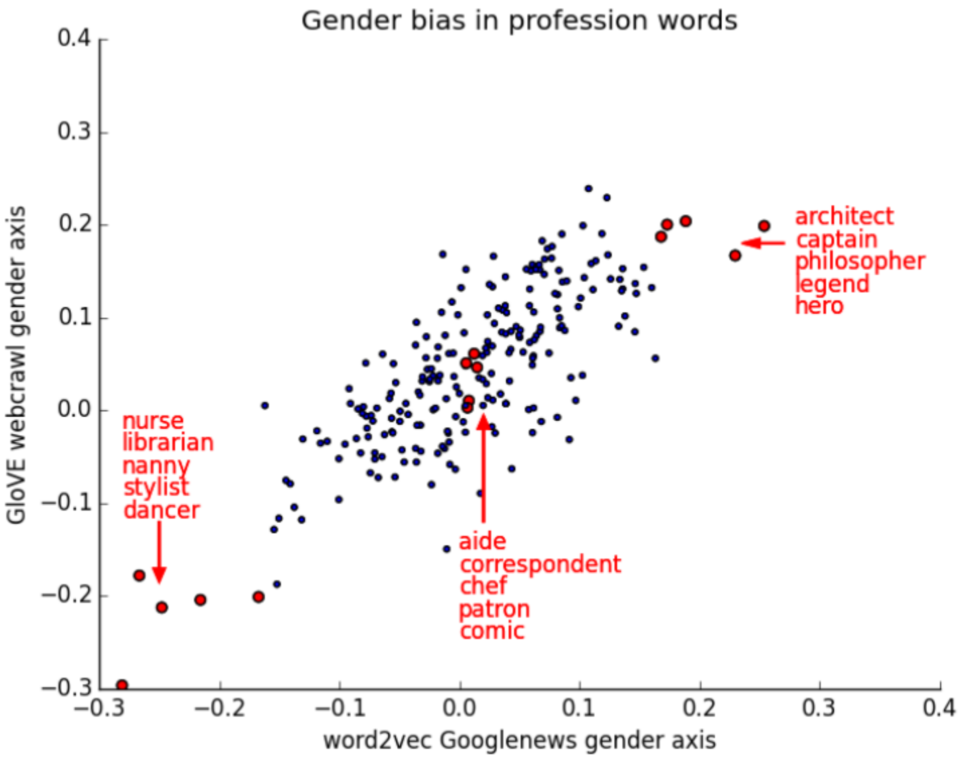
Representational Bias

Embeddings are *highly* sexists!

Bolukbasi, T., Chang, K.W., Zou, J., Saligrama, V. and Kalai, A., 2016.

Quantifying and reducing stereotypes in word embeddings. *arXiv preprint*





The European Union's draft AI regulations classify AI systems into three risk categories.

Unacceptable-risk AI systems	High-risk AI systems	Limited- and minimal-risk AI systems
• Subliminal, manipulative, or exploitative techniques causing harm • Real-time, remote biometric identification systems used in public spaces for law enforcement • All forms of social scoring	• Systems that evaluate consumer creditworthiness • Recruiting or employee-management systems • Systems utilizing biometric identification in nonpublic spaces • Safety-critical systems (eg, systems that would put the health of citizens at risk due to failure) • Any systems used in the administration of justice	• AI chatbots • AI-enabled video and computer games • Spam filters • Inventory-management systems • Customer- and market-segmentation systems • Most other AI systems

The draft EU AI regulations place several requirements on organizations providing or using high-risk AI systems.

- Implementation of a risk-management system
- Data governance and management
- Technical documentation
- Record keeping and logging
- Transparency and provision of information to users
- Human oversight
- Accuracy, robustness, and cybersecurity
- Conformity assessment
- Registration with EU-member-state government
- Postmarket monitoring system

https://www.mckinsey.com/business-functions/mckinsey-analytics/our-insights/what-the-draft-european-union-ai-regulations-mean-for-business

https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1623335154975&uri=CELEX%3A52021PC0206

ARTIFICIAL INTELLIGENCE

AI

EUROPEAN COMMISSION

Brussels, 21.4.2021
COM(2021) 206 final
2021/0106(COD)

Proposal for a

REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL

LAYING DOWN HARMONISED RULES ON ARTIFICIAL INTELLIGENCE (ARTIFICIAL INTELLIGENCE ACT) AND AMENDING CERTAIN UNION LEGISLATIVE ACTS

{SEC(2021) 167 final} - {SWD(2021) 84 final} - {SWD(2021) 85 final}

EXPLANATORY MEMORANDUM

1. CONTEXT OF THE PROPOSAL

1.1. Reasons for and objectives of the proposal

This explanatory memorandum accompanies the proposal for a Regulation laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Artificial Intelligence (AI) is a fast evolving family of technologies that can bring a wide array of economic and societal benefits across the entire spectrum of industries and social activities. By improving prediction, optimising operations and resource allocation, and personalising service delivery, the use of artificial intelligence can support socially and environmentally beneficial outcomes and provide key competitive advantages to companies and the European economy. Such action is especially needed in high-impact sectors, including climate change, environment and health, the public sector, finance, mobility, home affairs and agriculture. However, the same elements and techniques that power the socio-economic benefits of AI can also bring about new risks or negative consequences for individuals or the society. In light of the speed of technological change and possible challenges, the EU is committed to strive for a balanced approach. It is in the Union interest to preserve the EU's technological leadership and to ensure that Europeans can benefit from new technologies developed and functioning according to Union values, fundamental rights and principles.

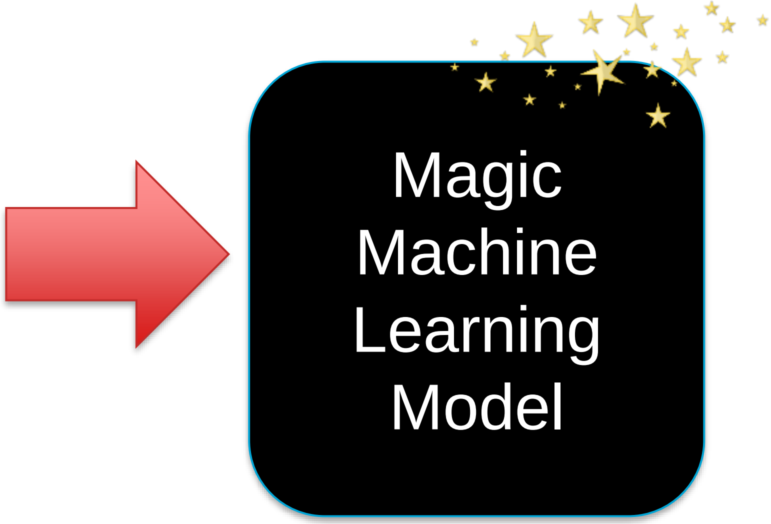
This proposal delivers on the political commitment by President von der Leyen, who announced in her political guidelines for the 2019-2024 Commission "A Union that strives for more" [1], that the Commission would put forward legislation for a coordinated European approach on the human and ethical implications of AI. Following on that announcement, on 19 February 2020 the Commission published the White Paper on AI - A European approach to excellence and trust [2]. The White Paper sets out policy options on how to achieve the twin objective of promoting the uptake of AI and of addressing the risks associated with certain uses of such technology. This proposal aims to implement the second objective for the development of an ecosystem of trust by proposing a legal framework for trustworthy AI. The proposal is based on EU values and fundamental rights

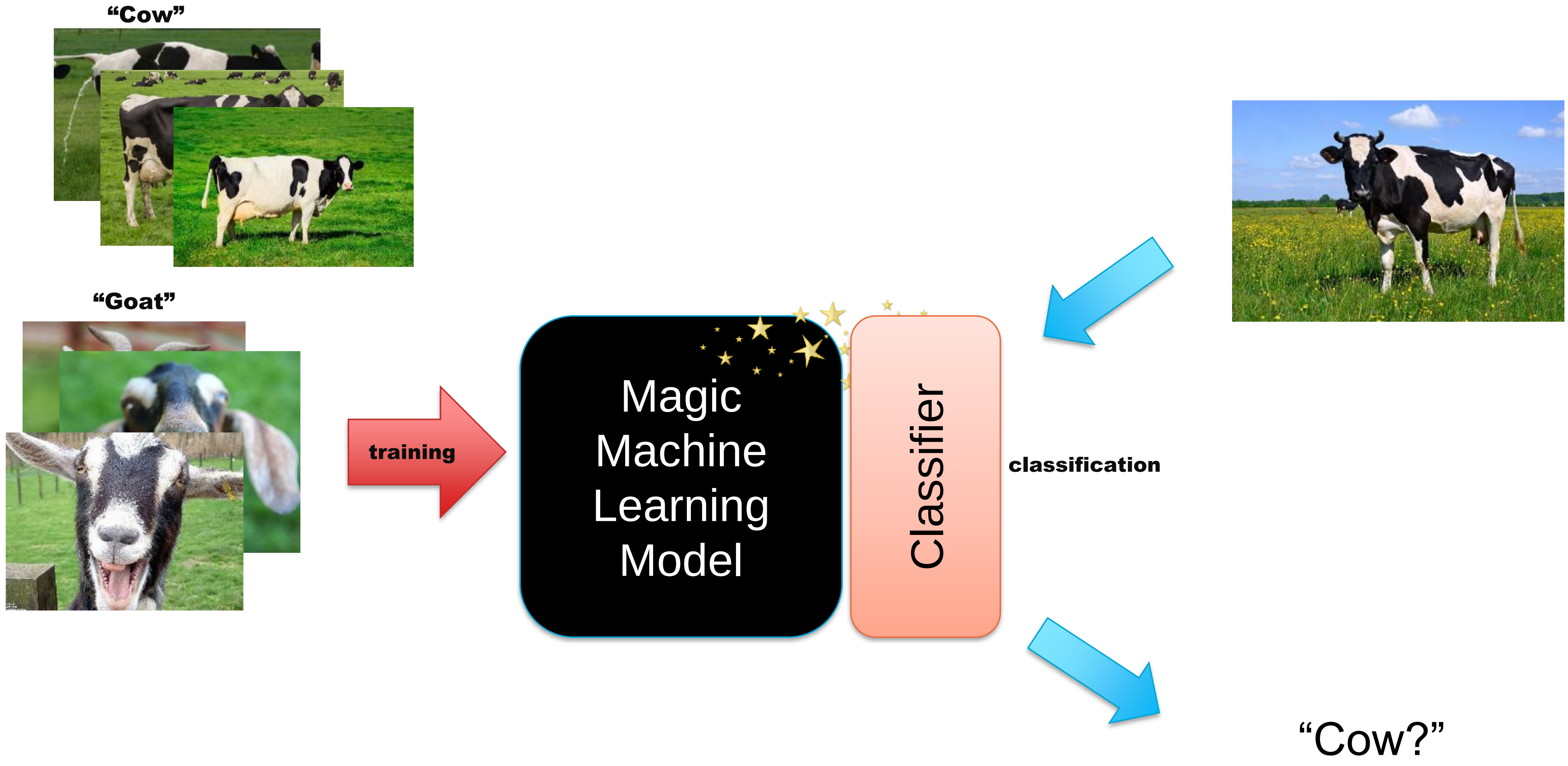
The Larger Issue?

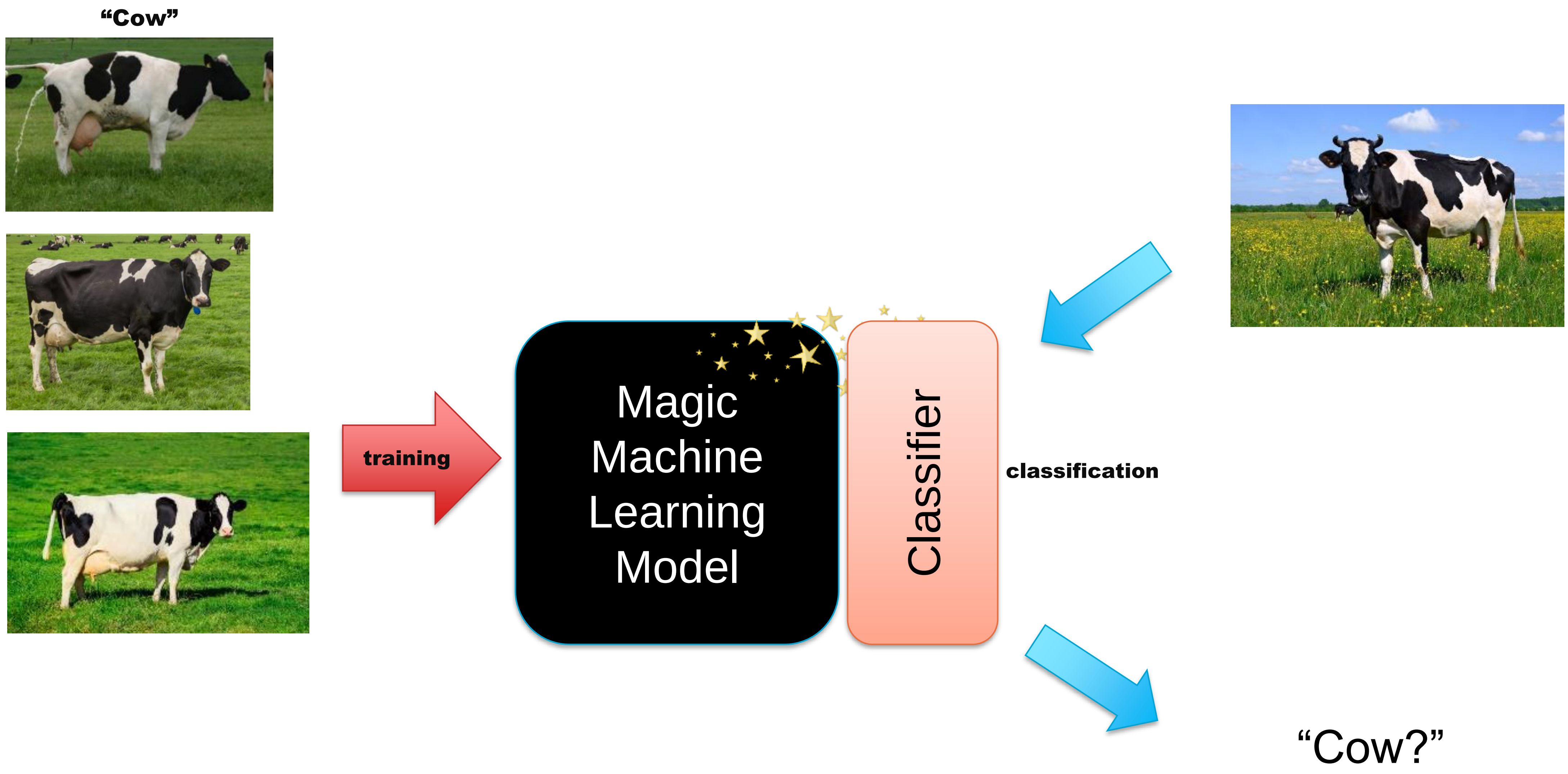
- It’s not just about bias, fairness and discrimination!

“Cow”

“Goat”




"Cow"
"Goat"
training
Magic
Machine
Learning
Model
Classifier
classification
"Cow?"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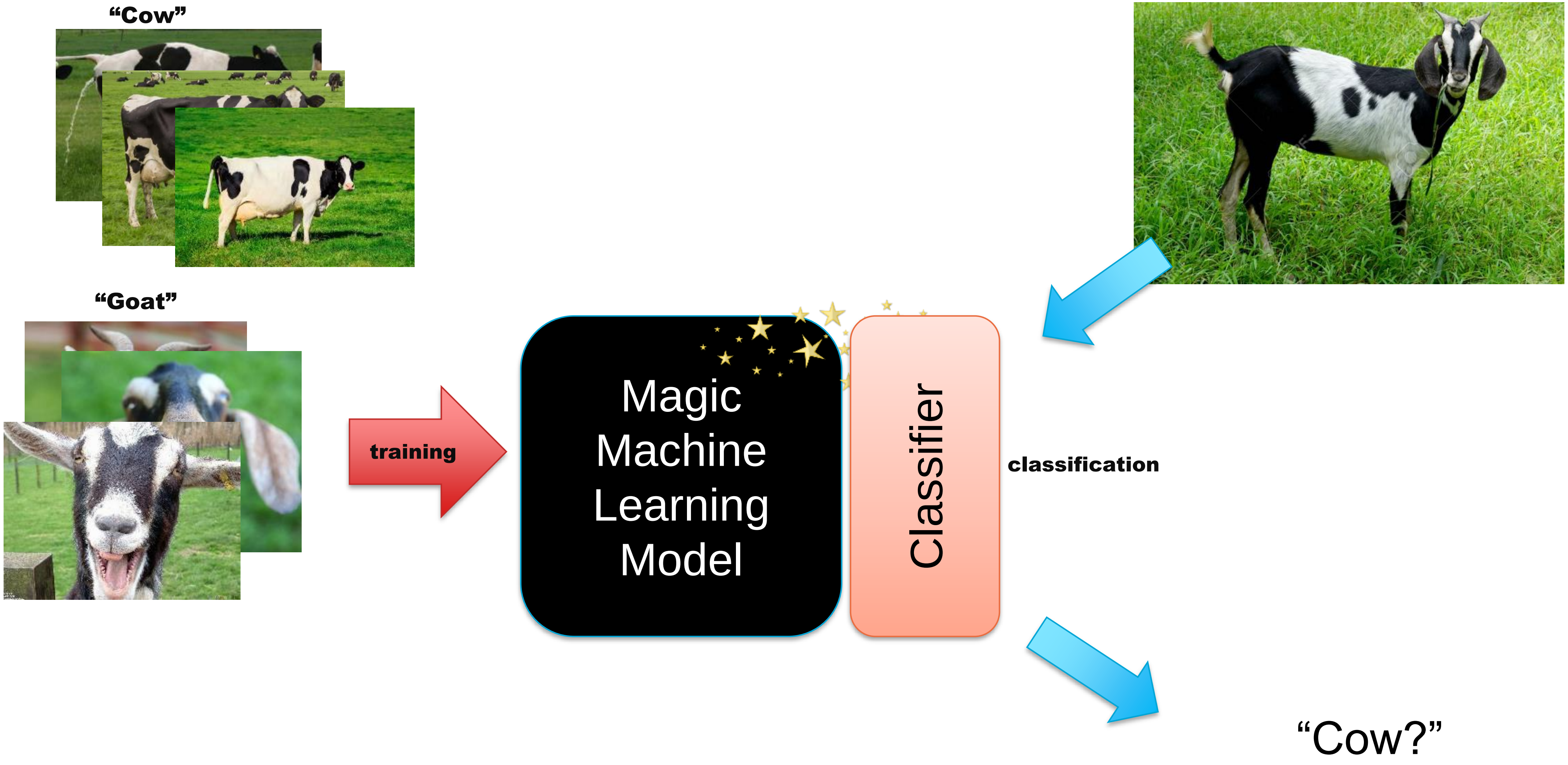
"Cow?"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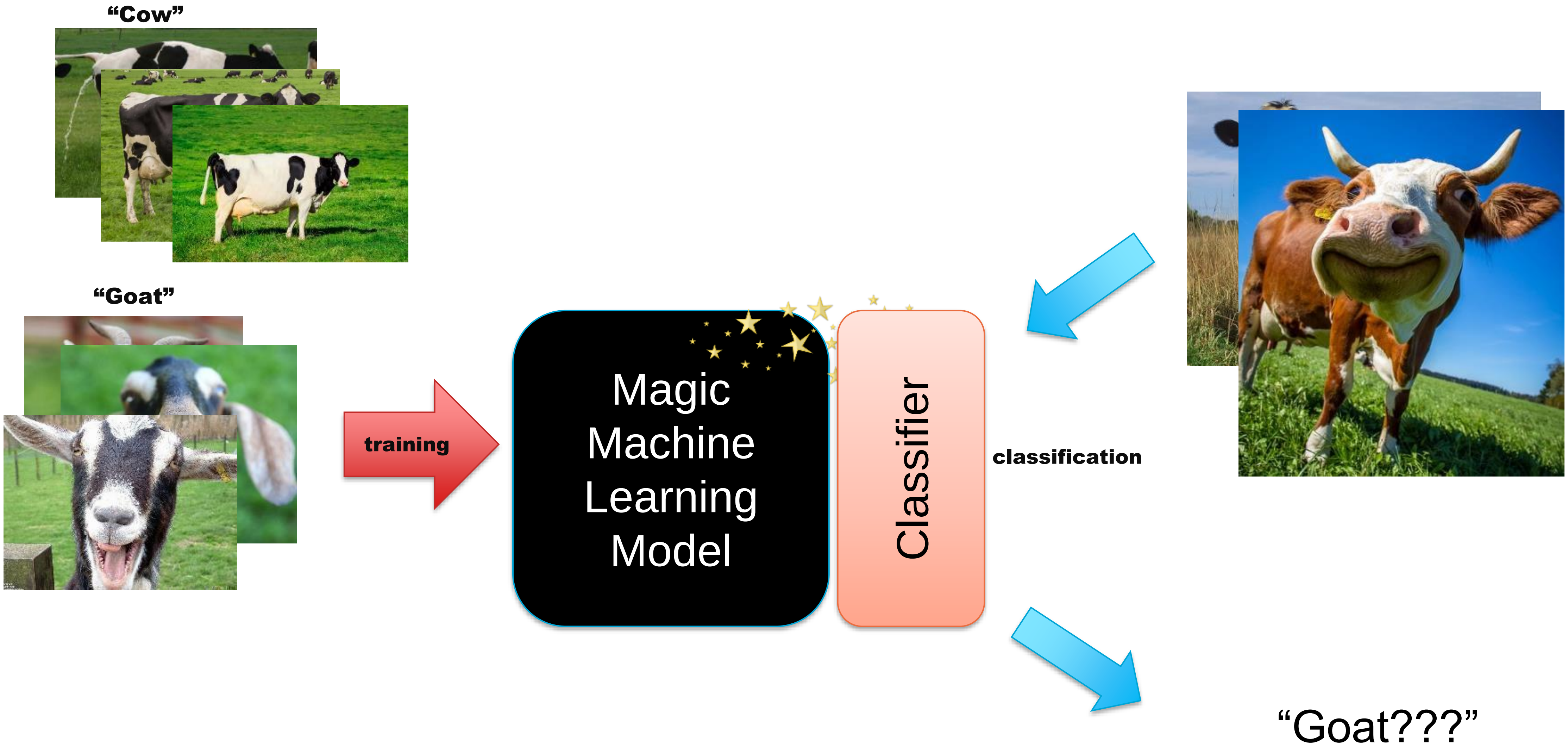

"Cow"
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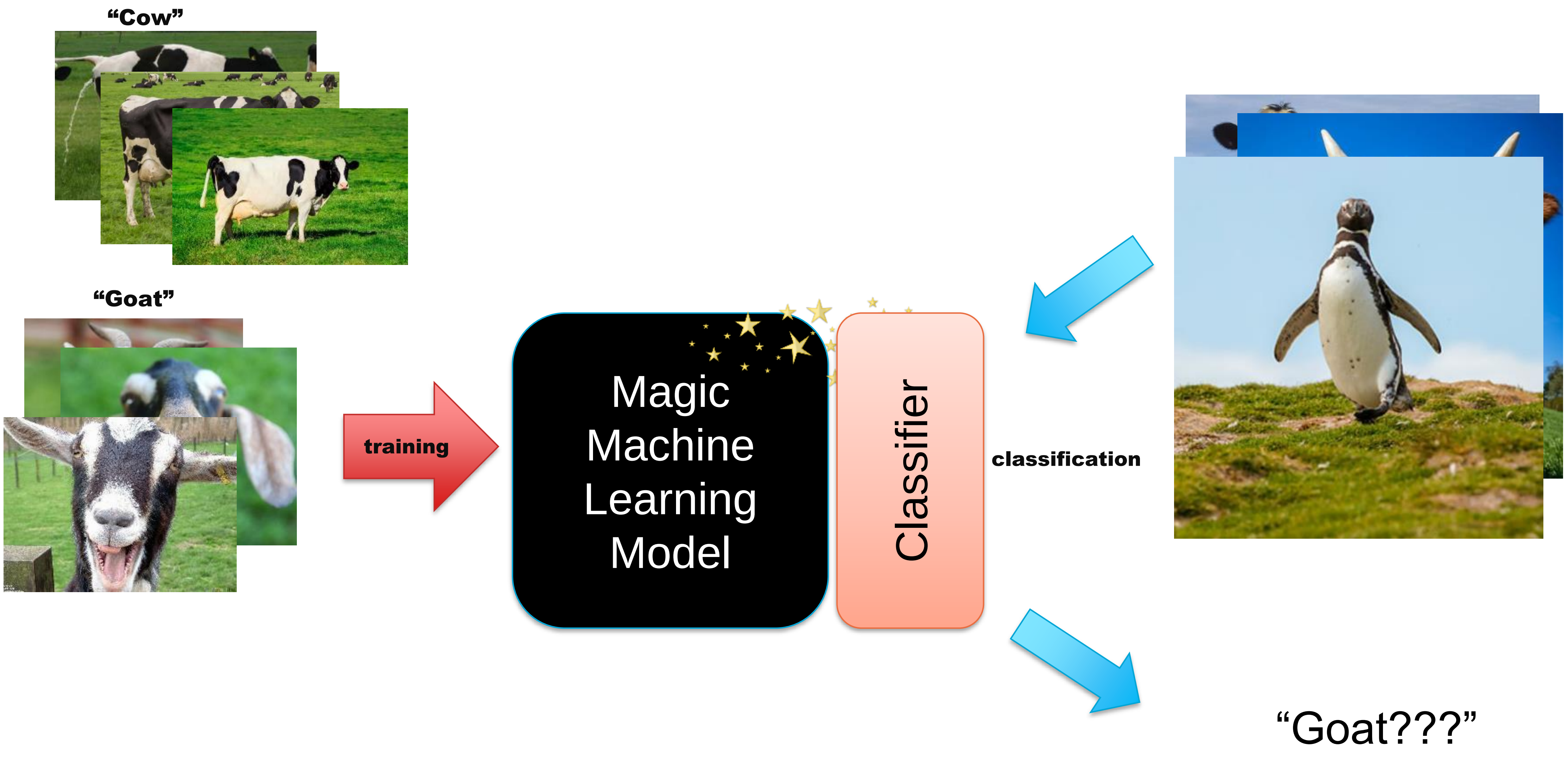
"Cow"
"Goat"
training
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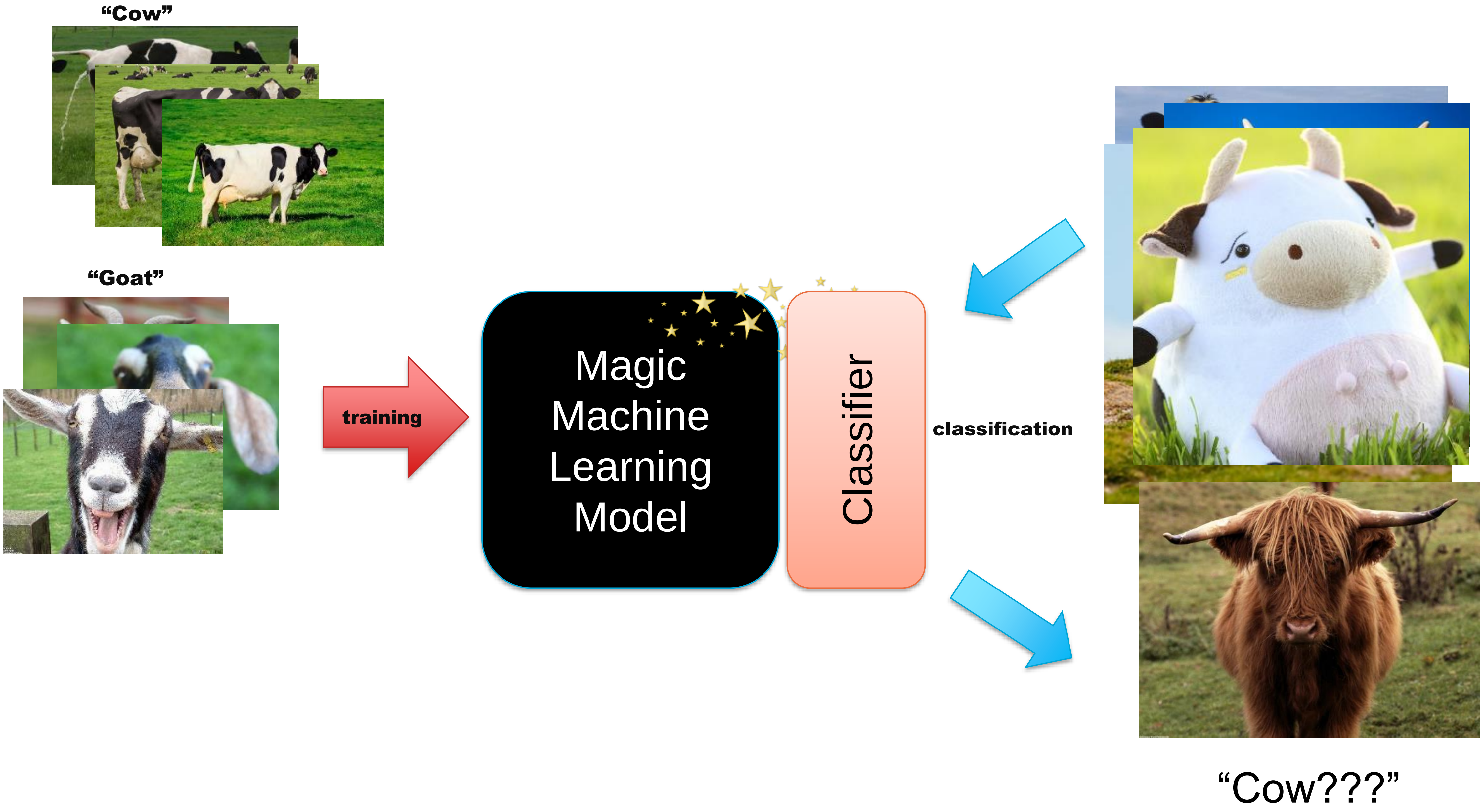
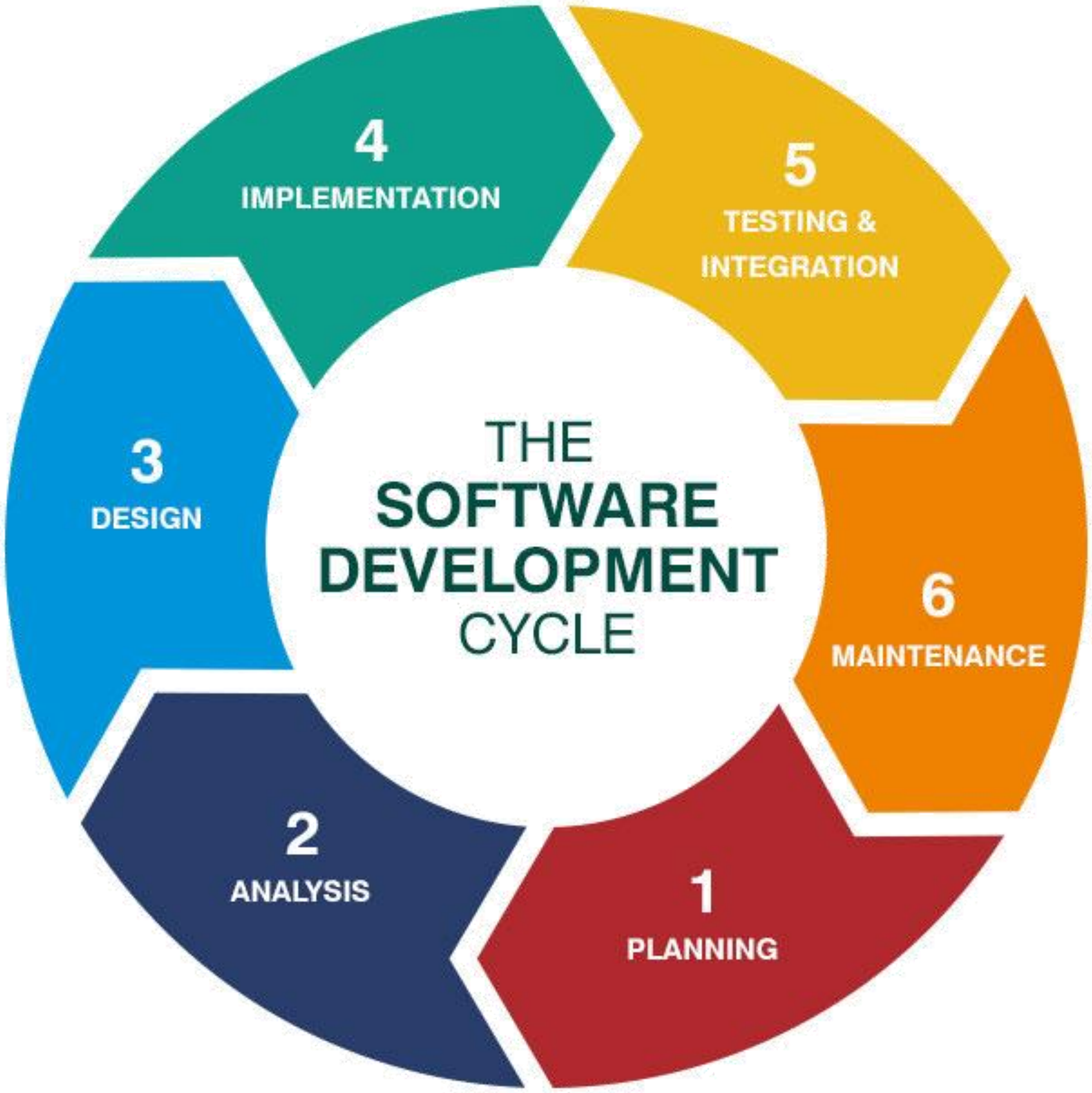
classification
"Goat???"

TRAINING DATA




THE SOFTWARE DEVELOPMENT CYCLE
1 PLANNING
2 ANALYSIS
3 DESIGN
4 IMPLEMENTATION
5 TESTING & INTEGRATION
6 MAINTENANCE

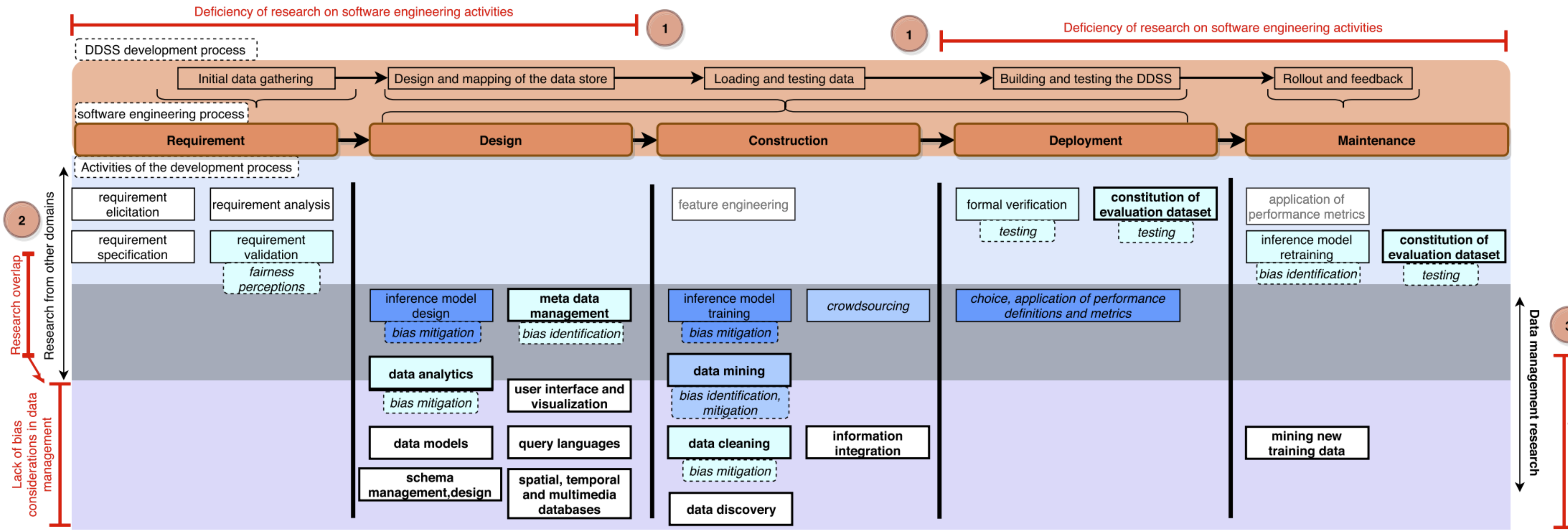


Fig. 2 Activities relevant to bias and their amount of bias-related research (white: no research; light to dark blue from few to plenty of researc
Data management activities are bold

A.M.A. Balayn, C. Lofi, G.J.P.M. Houben (2021), [Managing bias and unfairness in data for decision support: a survey of machine learning and data engineering approaches to identify and mitigate bias and unfairness within data management and analytics systems](), In ***The VLDB Journal*** **Volume** 30 p.739-768.

Bias-Aware Data Requirements

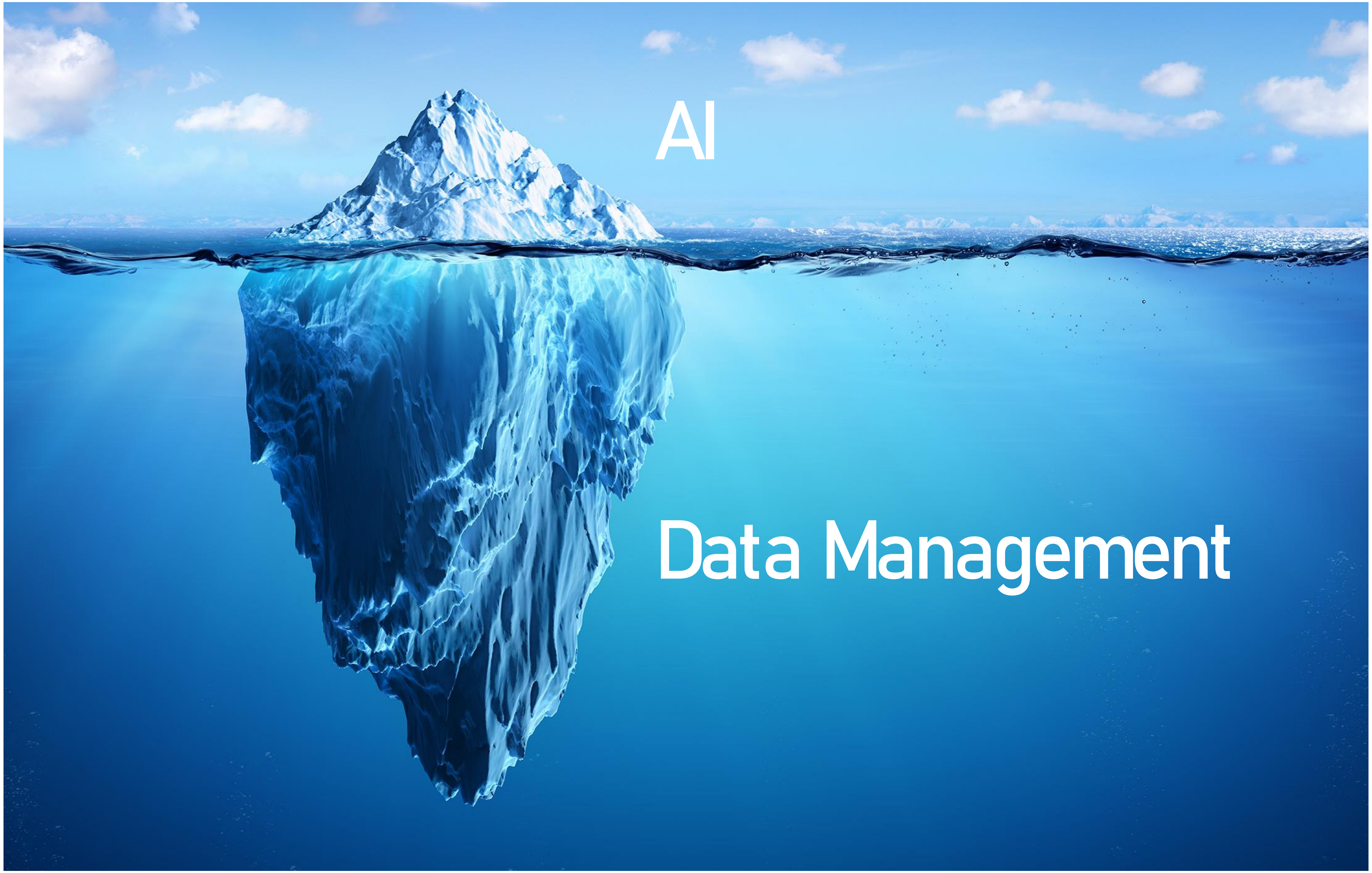
Bias Metrics

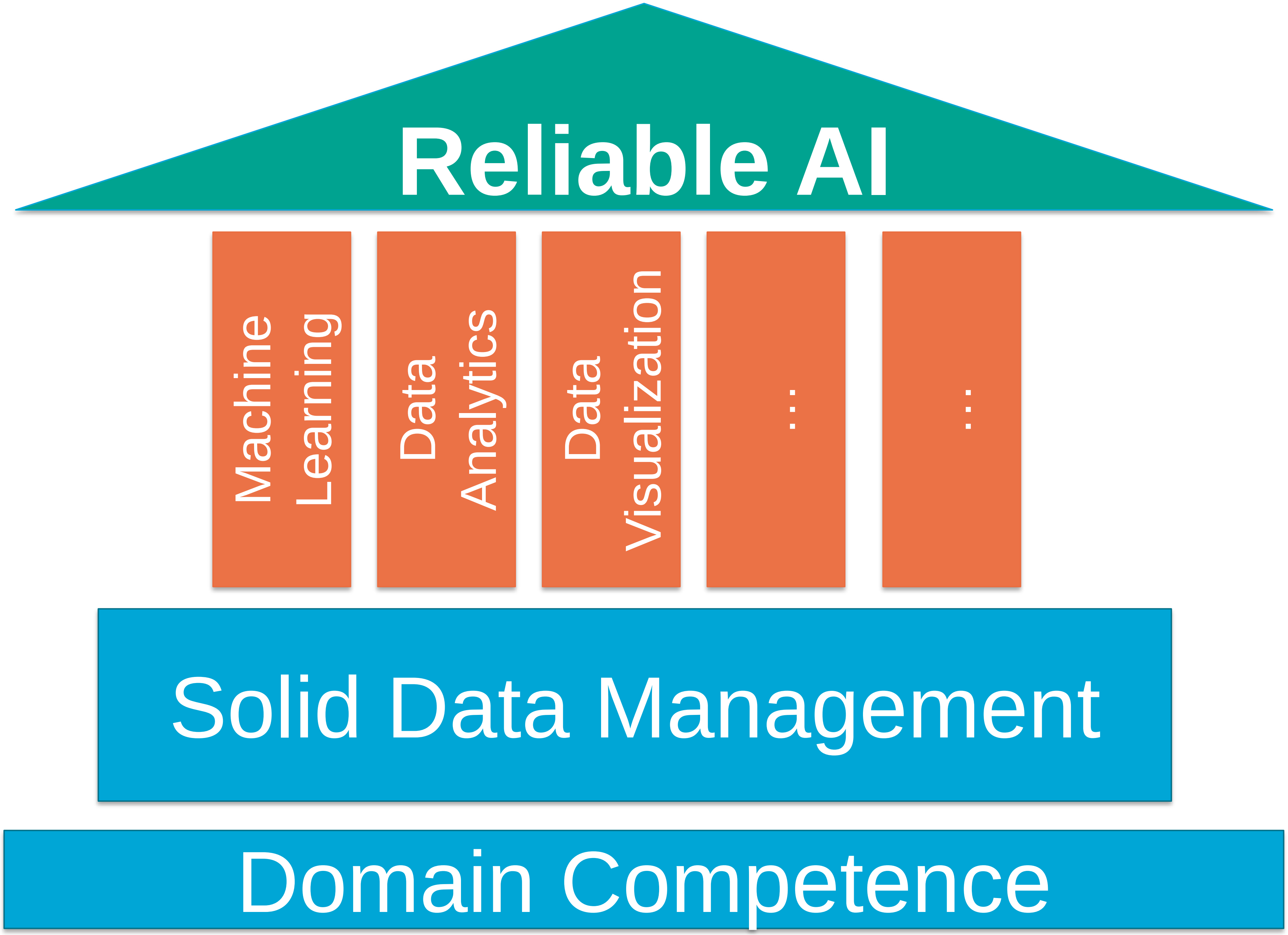
Bias-Aware Schema Design

Bias Constraints

Bias Curation Methods

Integration in Data Management & Engineering Systems?


AI
Data Management


Reliable AI
Machine Learning
Data Analytics
Data Visualization
…
…
Solid Data Management
Domain Competence

THANKS!

Christoph Lofi
c.lofi@tudelft.nl

TU Delft WIS Web Information Systems